

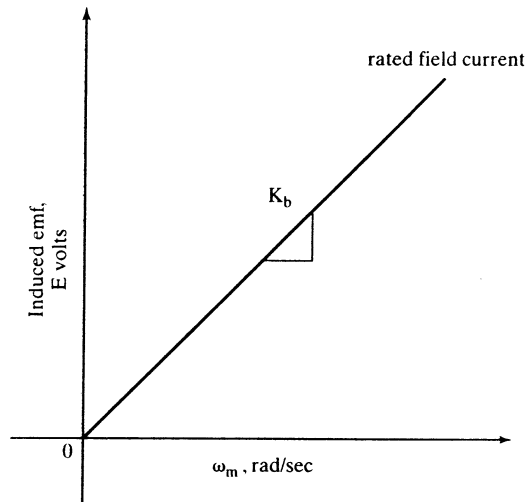


Figure 2.12 Induced emf-vs.-speed characteristic at rated field current

a voltmeter connected across the terminals. The voltmeter reads the induced emf, and its readings are noted for various speeds and are plotted as shown in Figure 2.12. The slope of this curve at a specific speed gives the emf constant in volt-sec/rad as seen from equation (2.4). The relationship shown in Figure 2.12 is known as the open-circuit characteristic of the dc machine.

Similar procedures for the field-circuit parameters, given in sections 2.8.1 and 2.8.2, are used to evaluate the resistance and inductance of the field circuit. Note that, for a permanent-magnet machine, this procedure is not applicable.

2.9 FLOW CHART FOR COMPUTATION

The basic steps involved in the computation of dc motor response and other quantities of interest are as follows:

- (i) Reading of machine parameters
- (ii) Specification of computational needs, such as time response, frequency response, eigenvalues for stability
- (iii) Use of transfer functions to evaluate the responses
- (iv) Printing/displaying of the output results

An illustrative flow chart containing these steps for a separately-excited dc motor is shown in Figure 2.13.

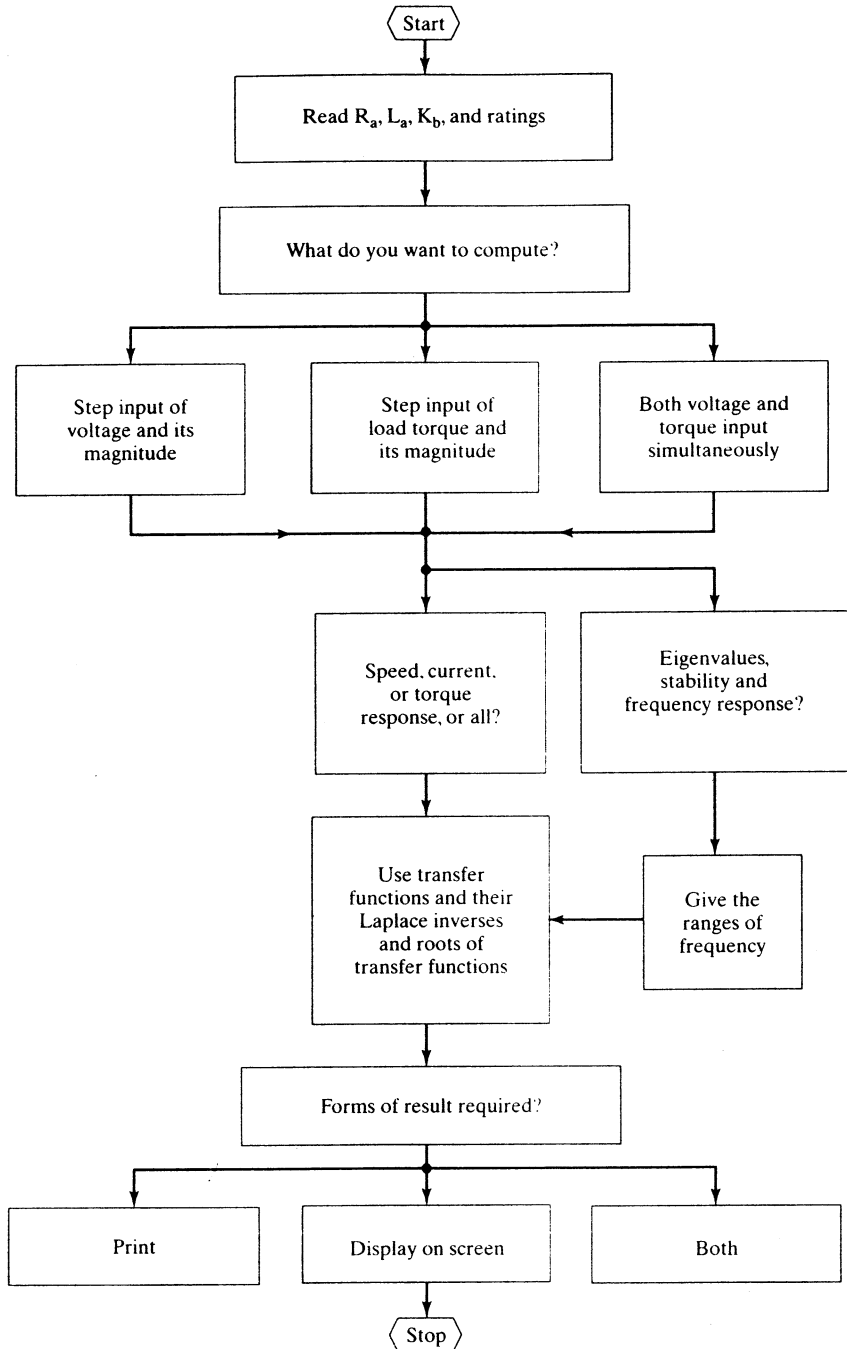


Figure 2.13 Flow chart for the computation of dc motor response

2.10 SUGGESTED READINGS

1. Fitzgerald, Kingsley, and Kusko, *Electric Machinery*, McGraw-Hill, 1971.
2. B.C. Kuo, *Automatic Control Systems*, Prentice-Hall, New York, 1982.

2.11 DISCUSSION QUESTIONS

1. A separately-excited dc motor has a nonlinear characteristic between the field current and field flux due to the saturation of the iron core in the stator and rotor. How will this saturation affect the derived model of the dc motor?
2. How is one to account for saturation in dc-machine modeling?
3. The armature resistance of the dc motor is sensitive to temperature variations. Will this adversely affect the stability of the dc motor?

2.12 EXERCISE PROBLEMS

1. Determine the torque vs. speed and torque vs. current characteristics for a separately-excited dc motor with the following parameters:

2.3 hp, 220 V, 6000 rpm, $R_a = 1.39 \Omega$, $L_a = 0.00182$ H, $K_b = 0.331$ volt/rad/sec.

The machine has rated field excitation and its armature is fed a constant voltage of 220 V dc.

2. Derive the dynamic equations of a dc series motor and find its characteristic equation. (Hint: To find the characteristic equation, perturb the dynamic equations to get a set of small-signal dynamic equations, and then find the characteristic equation, or directly calculate the eigenvalues of the A matrix.)
3. The machine given in 2.12.1 has $J = 0.002$ kg·m², $B = 0.005$ N·m/rad/sec. Determine the time taken to accelerate the motor from standstill to 6000 rpm when started directly from a 220-V dc supply. The field is maintained at its rated value.

4. The field current of a separately-excited motor is variable from zero to rated value. Derive the dynamic equations of the dc motor, its block diagram, and the small-signal armature-current response for a simultaneous voltage input to the field and torque disturbance, keeping the voltage applied to the armature constant. The motor is running at 1000 rpm, delivering 10 N·m torque, with half the rated flux. The machine details are given below:

400 V dc, 22.75 hp, 3600 rpm, $R_a = 0.34 \Omega$, $J = 0.035$ kg·m², $L_a = 1.13$ mH, $B_f = 0$ N·m/rad/sec, $K_b = 1.061$ volt/rad/sec, $M = 0.2122$ H, $R_f = 20 \Omega$, $L_f = 17.7$ H, $\Delta V_f = 5$ V, $\Delta T_f = 1$ N·m.

5. Using a computer program, find the transfer function between rotor speed and load torque and plot its frequency response. The separately-excited dc motor's details are as follows:

$R_a = 0.027 \Omega$, $L_a = 0.9$ mH, $K_b = 0.877$ volt/rad/sec, $J = 0.29$ kg·m², $B = 0.1$ N·m/rad/sec.

6. Determine the stability of a dc series motor with the following parameters:

$R_a = 1.5 \Omega$, $R_{se} = 0.7 \Omega$, $L_a = 0.12$ H, $L_{se} = 0.03$ H, $M = 0.0675$ H, $J = 0.02365$ kg·m², $B_f = 0.0025$ N·m/rad/sec, and the operating point is given by $V_a = 200$ V, $\omega_m = 209.52$ rad/sec, and $T_f = 10$ N·m.

CHAPTER 3

Phase-Controlled DC Motor Drives

3.1 INTRODUCTION

The principle of speed control for dc motors is developed from the basic emf equation of the motor. Torque, flux, current, induced emf, and speed are normalized to present the motor characteristics. Two types of control are available: armature control and field control. These methods are combined to yield a wide range of speed control. The torque–speed characteristics of the motor are discussed for both directions of rotation and delivering both motoring and regenerating torques in any direction of rotation. Such an operation, known as four-quadrant operation, has a unique set of requirements on the input voltage and current to the dc motor armature and field. These requirements are identified for specifying the power stage.

Modern power converters constitute the power stage for variable-speed dc drives. These power converters are chosen for a particular application depending on a number of factors such as cost, input power source, harmonics, power factor, noise, and speed of response. Controlled-bridge rectifiers fed from single-phase and three-phase ac supply are considered in this chapter. Chapter 4 deals with another converter, fed from a dc source, for dc motor control.

The theory, operation and control of the three-phase controlled-bridge rectifier is considered in detail, because of its widespread use. A model for the power converter is derived for use in simulation and controller design. Two- and four-quadrant dc motor drives and their control are developed. The design of the current and speed controllers is studied with an illustrative example. The interaction of the converter and motor is also discussed, and an illustrative example of their interaction with power system is presented. An industrial application of the motor drive is described.

3.2 PRINCIPLES OF DC MOTOR SPEED CONTROL

3.2.1 Fundamental Relationship

The dependence of induced voltage on the field flux and speed has been derived in Chapter 2 and is given as

$$e = K\phi_f \omega_m \quad (3.1)$$

The various symbols in equation (3.1) have been explained earlier. The field flux is proportional to the field current if the iron is not saturated and is represented as

$$\phi_f \propto i_f \quad (3.2)$$

By substituting (3.2) into (3.1) the speed is expressed as

$$\omega_m \propto \frac{e}{\phi_f} \propto \frac{e}{i_f} \propto \frac{(v - i_a R_a)}{i_f} \quad (3.3)$$

where v and i_a are the applied voltage and armature current, respectively.

From equation (3.3), it is seen that the rotor speed is dependent on the applied voltage and field current. Since the resistive armature voltage drop is very small compared to the rated applied voltage, the armature current has only a secondary effect. To make its effect dominant, an external resistor in series with armature can be connected. In that case, the speed can be controlled by varying stepwise the value of the external resistor as a function of operational speed. Power dissipation in the external resistor leads to lower efficiency; therefore, it is not considered in this text. Only two other forms of control, using armature voltage and field current, are considered in this chapter.

3.2.2 Field Control

In *field control*, the applied armature voltage v is maintained constant. Then the speed is represented by equation (3.3) as

$$\omega_m \propto \frac{1}{i_f} \quad (3.4)$$

The rotor speed is inversely proportional to the field current; by varying the field current, the rotor speed is changed. Reversing the field current changes the rotational direction. By weakening the field flux, the speed can be increased. The upper speed is limited by the commutator and brushes and the time required to turn off the armature current from a commutator segment. It is not possible to strengthen the field flux beyond its rated (nominal) value, on account of saturation of the steel laminations. Hence, field control for speed variation is not suitable below the rated (nominal) speed. At rated speed, the field current by design is at rated value, and, hence, the flux density is chosen to be near the knee of the magnetization curve of the steel laminations.

3.2.3 Armature Control

In this mode, the field current is maintained constant. Then the speed is derived from equation (3.3) as

$$\omega_m \propto (v - i_a R_a) \quad (3.5)$$

Hence, varying the applied voltage changes speed. Reversing the applied voltage changes the direction of rotation of the motor.

Armature control has the advantage of controlling the armature current swiftly, by adjusting the applied voltage. The response is determined by the armature time constant, which has a very low value. In contrast, the field time constant is at least 10 to 100 times greater than the armature time constant. The large time constant of the field causes the response of a field-controlled dc motor drive to be slow and sluggish.

Armature control is limited in speed by the limited magnitude of the available dc supply voltage and armature winding insulation. If the supply dc voltage is varied from zero to its nominal value, then the speed can be controlled from zero to nominal or rated value. Therefore, armature control is ideal for speeds lower than rated speed; field control is suitable above for speeds greater than the rated speed.

3.2.4 Armature and Field Control

By combining armature and field control for speeds below and above the rated speed, respectively, a wide range of speed control is possible. For speeds lower than that of the rated speed, applied armature voltage is varied while the field current is kept at its rated value; to obtain speeds above the rated speed, field current is decreased while keeping the applied armature voltage constant. The induced emf, power, electromagnetic torque, and field-current-vs.-speed characteristics are shown in Figure 3.1. The armature current is assumed to be equal to the rated value for the present. The power and torque curves need some elaboration. It can be deduced that

$$T_e = K \phi_f i_a \quad (3.6)$$

Equation (3.6) can be normalized if it is divided by rated torque, which is expressed as

$$T_{er} = K \phi_{fr} i_{ar} \quad (3.7)$$

where the additional subscript r denotes the rated or nominal values of the corresponding variables. Hence, the normalized version of equation (3.6) is

$$T_{en} = \frac{T_e}{T_{er}} = K \frac{\phi_f i_a}{\phi_{fr} i_{ar}} = \left(\frac{\phi_f}{\phi_{fr}} \right) \left(\frac{i_a}{i_{ar}} \right) = \phi_{fn} i_{an}, \text{ p.u.} \quad (3.8)$$

where the additional subscript n expresses the variables in normalized terms, commonly known as per unit (p.u.) variables.

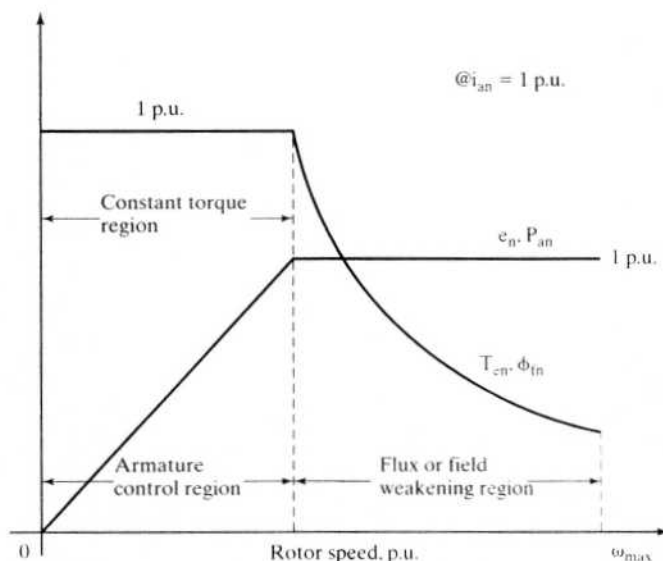


Figure 3.1 Normalized characteristics of a variable-speed dc motor

Normalization eliminates machine constants, compacts the performance equations, and enables the visualization of performance characteristics regardless of machine size on the same scale. Designers and seasoned analysts prefer the p.u. representation because of all these features. The normalized torque, flux, and armature current are

$$T_{cn} = \frac{T_c}{T_{cr}}, \text{ p.u.} \quad (3.9)$$

$$\phi_{fn} = \frac{\phi_f}{\phi_{fr}}, \text{ p.u.} \quad (3.10)$$

$$i_{an} = \frac{i_a}{i_{ar}}, \text{ p.u.} \quad (3.11)$$

As the armature current is maintained at 1 p.u. in Figure 3.1, the normalized torque becomes

$$T_{cn} = \phi_{fn}, \text{ p.u.} \quad (3.12)$$

Hence, the normalized electromagnetic torque characteristic coincides with the normalized field flux, as shown in Figure 3.1.

Similarly, the air gap power is,

$$P_{an} = e_n i_{an}, \text{ p.u.} \quad (3.13)$$

where e_n is the normalized induced emf.

As i_{an} is set to 1 p.u., the normalized air gap power becomes

$$P_{an} = e_n, \text{ p.u.} \quad (3.14)$$

The normalized power output characteristic is similar to the induced emf of the dc motor in the field-weakening and constant-torque regions. The normalized induced emf is the product of the normalized flux and speed. Flux is at 1 p.u. in the armature control region, so the normalized induced emf is equal to the normalized speed. The flux is hyperbolic in the field-weakening region and has an inverse relationship with the speed. Field weakening needs to be discussed in detail here. At rated speed, the motor is delivering rated power with e_n and i_{an} at their rated values. Beyond the rated speed, the field current is decreased to reduce the field flux. This will affect the magnitude of the induced emf and hence the power output. It is very important that the steady-state power output of the machine be kept from exceeding its rated design value, which is 1 p.u. The implication of the air gap power constraint is that the induced emf and field flux are to be coordinated to achieve this objective. The coordination yields the value of field flux as

$$P_{an} = 1 \text{ p.u.} = e_n i_{an} = \phi_{in} \omega_{mn} i_{an} \quad (3.15)$$

If i_{an} is equal to 1 p.u., then

$$\begin{aligned} \phi_{in} \omega_{mn} &= 1 \\ \phi_{in} &= \frac{1}{\omega_{mn}} \end{aligned} \quad (3.16)$$

Hence, the normalized induced emf is given as,

$$e_n = \phi_{in} \omega_{mn} = \frac{1}{\omega_{mn}} \times \omega_{mn} = 1 \text{ p.u.} \quad (3.17)$$

The power output and induced emf are maintained at their rated values in the field-weakening region by programming the field flux to be inversely proportional to the rotor speed. They are shown in Figure 3.1.

Example 3.1

A separately-excited dc motor has the following ratings and constants:

2.625 hp., 120 V, 1313 rpm, $R_a = 0.8 \, \Omega$, $R_f = 100 \, \Omega$, $K_b = 0.764 \text{ V.s / rad}$, $L_a = 0.003 \text{ H}$, $L_f = 2.2 \text{ H}$

The dc supply voltage is variable from 0 to 120 V both to the field and armature, independently. Draw the torque-speed characteristics of the dc motor if the armature and field currents are not allowed to exceed their rated values. The rated flux is obtained when the field voltage is 120 V. Assume that the field voltage can be safely taken to a minimum of 12 V only.

Solution (i) Calculation of rated values:

$$\text{Rated speed, } \omega_{mr} = \frac{2\pi N}{60} = \frac{2\pi \times 1313}{60} = 137.56 \text{ rad/sec}$$

$$\text{Rated torque, } T_{er} = \frac{\text{Output power}}{\text{Rated speed}} = \frac{2.625 \times 745.6}{137.56} = 14.23 \text{ N}\cdot\text{m}$$

$$\text{Rated armature current, } I_{ar} = \frac{\text{Rated torque}}{K_b} = \frac{14.23}{0.764} = 18.63 \text{ A}$$

$$\text{Rated field current, } I_{fr} = \frac{V_{fr}}{R_f} = \frac{120}{100} = 1.2 \text{ A}$$

(ii) Calculation of torque-speed characteristics:

Case (a) Constant-flux/torque region:

$$e_1 = V_{\max} - I_{ar} R_a = 120 - 18.63 \times 0.8 = 105.1$$

$$\omega_{m1} = \frac{e_{m1}}{K_b} = \frac{105.1}{0.764} = 137.56 \text{ rad/sec.}$$

$$\omega_{m1n} = \frac{\omega_{m1}}{\omega_{mr}} = \frac{137.56}{137.56} = 1.0 \text{ p.u.}$$

Hence, constant rated torque is available from 0 to 1.0 p.u. speed.

Case (b) Field-weakening region:

For 1 p.u. armature current, the maximum induced emf is

$$e_n = \frac{e_1}{e_r} = \frac{105.1}{105.1} = 1.0 \text{ p.u.}$$

To maintain this induced emf in the field-weakening region.

$$\phi_{fn} = \frac{e_n}{\omega_{mn}} = \frac{1.0}{\omega_{mn}} \text{ p.u.}$$

If the range of field variation is known, the maximum speed can be computed as follows:

$$I_{f \min} = \frac{V_{f(\min)}}{R_f} = \frac{12}{100} = 0.12 \text{ A}$$

1.2 A of field current corresponds to rated field flux and hence 0.12 A corresponds to $0.1\phi_{fr}$, and hence

$$0.1 \text{ p.u.} < \phi_{fn} < 1 \text{ p.u.}$$

$$\omega_{\max} = \frac{1}{0.1} = 10 \text{ p.u.}$$

For various speeds between 1 and 10 p.u., the field flux is evaluated from the equation as

$$\phi_{fn} = \frac{1}{\omega_{mn}} \text{ in p.u.}$$

$$T_{en} = \phi_{fn} \text{ for } I_{an} = 1 \text{ p.u.}$$

The torque, power, and flux-vs.-speed plots are shown in Figure 3.2.

Example 3.2

Consider the dc motor given in Example 3.1, and draw the intermittent characteristics if the armature current is allowed to be 300% of rated value.

Solution (i) Constant-flux/torque region

$$I_{\max} = 3I_{ar}$$

$$T_{em} = \text{maximum torque} = K_b I_{\max} = 0.764 \times 3 \times 18.63 = 42.7 \text{ N}\cdot\text{m.}$$

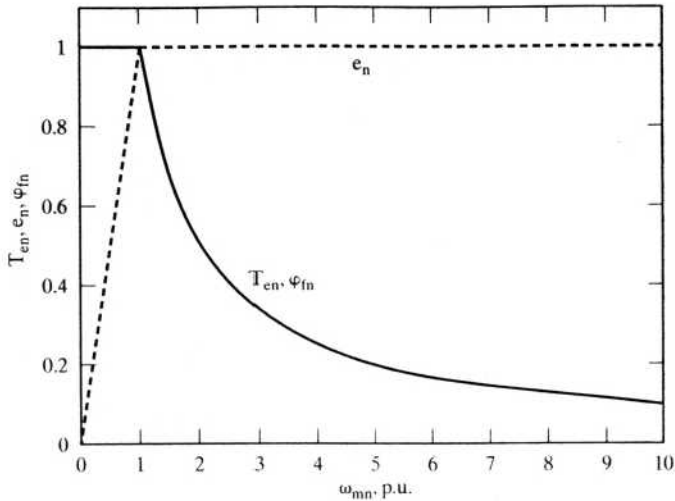


Figure 3.2 Continuous rating of the dc motor

$$T_{en} = \frac{T_{em}}{T_{er}} = \frac{42.7}{14.25} = 3 \text{ p.u.}$$

The maximum induced emf is

$$e_m = V_{\max} - I_{\max}R_a = 120 - (3 \times 18.63) \times 0.8 = 75.29 \text{ V}$$

Speed corresponding to this induced emf is

$$\omega_{ml} = \frac{e_m}{K_b} = \frac{75.29}{0.764} = 98.54 \text{ rad/sec}$$

$$\omega_{mln} = \frac{98.54}{137.56} = 0.716 \text{ p.u.}$$

Beyond this speed, field weakening is performed.

(ii) Field-weakening region:

$$I_{\max} = 3I_{ar}$$

$$e_m = 75.29 \text{ V}$$

$$e_n = \frac{e_m}{105.1} = \frac{75.29}{105.1} = 0.716 \text{ p.u.}$$

$$\omega_{mn} = \frac{e_n}{\phi_{fn}} = \frac{0.716}{\phi_{fn}} \text{ p.u.}$$

The range of the normalized field flux is

$$0.1 < \phi_{fn} < 1$$

The maximum normalized speed is $\omega_{mn} = \frac{0.716}{\phi_{fn \min}} = \frac{0.716}{0.1} = 7.16 \text{ p.u.}$

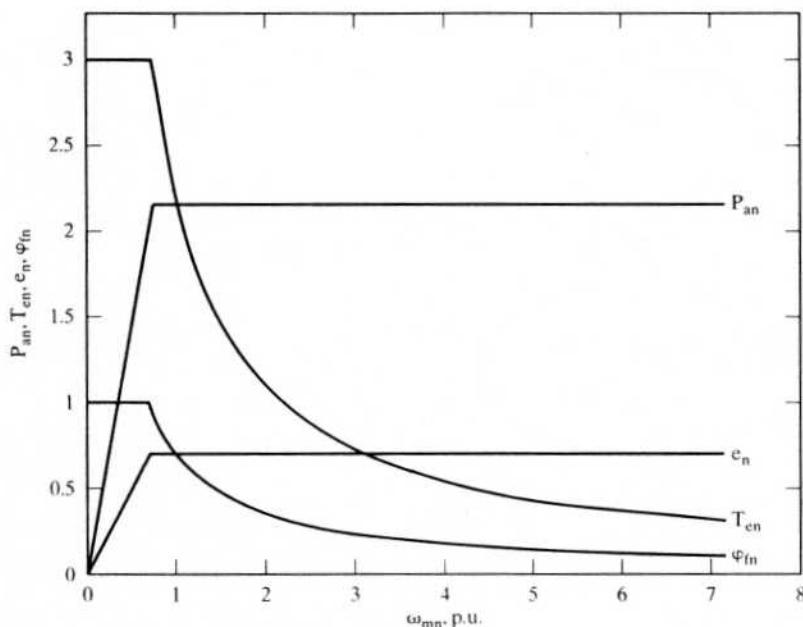


Figure 3.3 Normalized motor characteristics for 3 p.u. armature current

$$T_{en} = \phi_{fn} \text{ per rated current} = 3\phi_{fn} \text{ in the present case.}$$

The intermittent characteristics are drawn from the above equations and are shown in Figure 3.3.

3.2.5 Four-Quadrant Operation

Many applications require controlled starts and stops of the dc motor, such as in robotic actuation. Consider that the machine is operating at a steady speed of ω_m , and it is desired to bring the speed to zero. There are two ways to achieve it:

1. Cut off the armature supply to the machine and let the rotor come to zero speed.
2. The machine can be made to work as a dc generator, thereby the stored kinetic energy can be effectively transferred to the source. This saves energy and brings the machine rapidly to zero speed.

Cutting off supply produces a haphazard speed response; the second method provides a controlled braking of the dc machine. To make the dc machine operating in the motoring mode go to the generating mode, all that needs to be done is to reverse the armature current flow in the dc machine. First, the armature current drawn from the source has to be brought to zero; then, a current in the opposite direction has to be built. Zeroing the current is achieved by making the source dc voltage zero or, better, by making it negative. After this, the armature current is built in the opposite direction by making the source voltage smaller than the induced

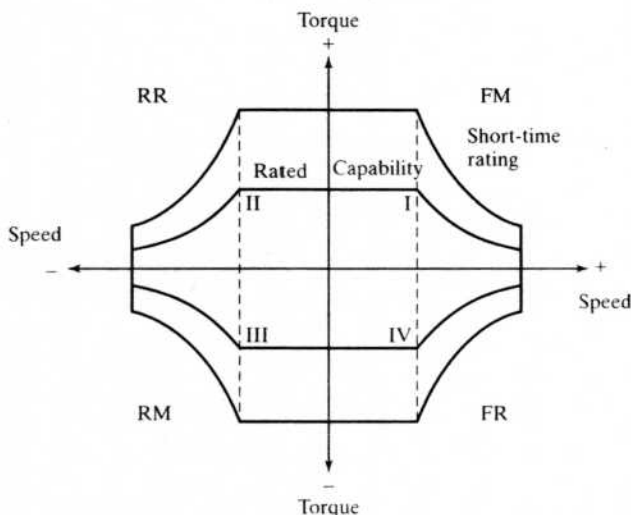


Figure 3.4 Four-quadrant torque-speed characteristics

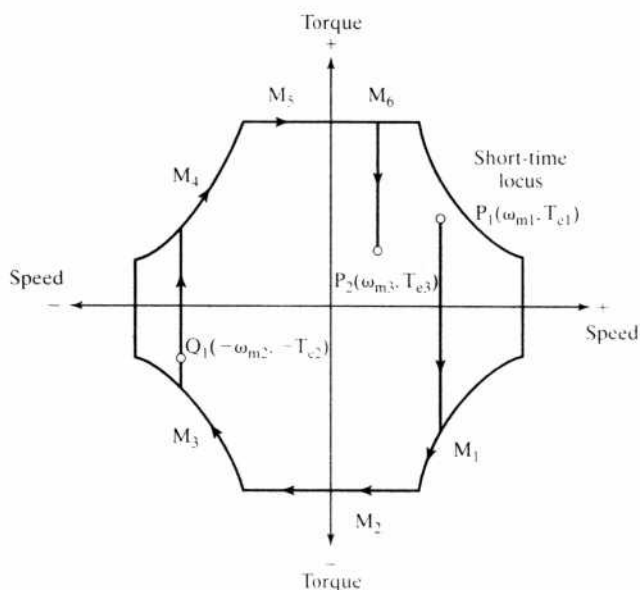
emf. As the speed reduces, note that the induced emf decreases, necessitating a continued corresponding reduction in the source voltage to keep the armature current constant. The power flows from the machine armature to the dc source. This mode of operation is termed *regenerative braking*. The braking is accomplished by regeneration that implies that a negative torque is generated in the machine as opposed to the positive motoring torque. Hence, a mirror reflection of the speed-torque characteristics, shown in Figure 3.1, is required on the IV quadrant for regeneration. The first and fourth quadrants are for one direction of rotation, say, forward.

Some applications, such as a feed drive in machine tools, require operation in both directions of rotation. In that case, the III quadrant signifies the reverse motoring and II quadrant, the reverse regeneration mode. A motor drive capable of operating in both directions of rotation and of producing both motoring and regeneration is referred to as a four-quadrant variable-speed drive. The torque-speed characteristics of such a four-quadrant dc motor drive are shown in Figure 3.4. This contains two characteristics, one for rated operating condition and the other for short-time or intermittent operation. The short-time characteristic is used for acceleration and deceleration of the machine; it normally encompasses 50 to 100% greater than the rated torque for dc machines. The four-quadrant operation and its relationship to speed, torque, and power output are summarized in Table 3.1.

Figure 3.5 illustrates the speed and torque variation from a point P_1 to Q_1 and Q_1 to P_2 of the dc machine. On receiving the command to go from P_1 to Q_1 , the torque is changed to negative by regenerating the machine, as shown by the trajectory P_1M_1 . This regeneration torque, along with the load torque, produces a decelerating torque. The torque is maintained at the permitted maximum levels both in the field-weakening and the constant-flux regions. As the machine decelerates, as shown by the trajectory M_1M_2 , it will reach zero speed, and keeping the torque at a negative maximum will drive the motor in the reverse direction along the trajectory M_2M_3 . Once the desired speed ω_{m2} is reached, the torque is adjusted to equal the

TABLE 3.1 Four-quadrant dc motor drive characteristics

Function	Quadrant	Speed	Torque	Power Output
Forward Motoring (FM)	I	+	+	+
Forward Regeneration (FR)	IV	+	-	-
Reverse Motoring (RM)	III	-	-	+
Reverse Regeneration (RR)	II	-	+	-

**Figure 3.5** Changing the operating points and the use of four quadrants

specified value, $-T_{e2}$, along the trajectory M_3Q_1 . Similarly, to change the operating point from Q_1 to P_2 , the trajectory shown along $Q_1M_4M_5M_6P_2$ is followed.

From this illustration, it is seen that the use of all quadrants of operation leads to a very responsive motor drive. Contrast this to the *supply cut-off* technique. In such a case, only the load torque contributes to the deceleration, as opposed to the combined machine and load torques in a four-quadrant motor drive.

Converter requirements: The voltage and current requirements for four-quadrant operation of the dc machine are derived as follows. Assuming the field flux is constant, the speed is proportional to the induced emf and hence approximately proportional to the applied voltage to the armature. Also, the electromagnetic torque is proportional to the armature current. Then the speed axis becomes the armature voltage axis and the torque axis is equivalent to the armature current axis. From this observation, the armature voltage and armature current requirements for four-quadrant operation are given in Table 3.2. These requirements, in turn, set the specifications for the electronic converter. For a pump application, the motor needs only unidirectional operation with no regenerative braking. Hence, only first-quadrant operation

TABLE 3.2 Armature voltage and current requirements of a four-quadrant dc motor drive

Operation	Speed	Torque	Voltage	Current	Power Output
FM	+	+	+	+	+
FR	+	-	+	-	-
RM	-	-	-	-	+
RR	-	+	-	+	-

is required, thereby limiting the converter specification to only positive voltage and current variations. Therefore, the power flow is unidirectional from source to load. For a golf cart electric-vehicle propulsion-drive application, a four-quadrant operation is required, with the attendant converter capability to handle power in both directions with bipolar voltage and current requirements. Then the converter is much more complex than that required for a one-quadrant drive.

From a fixed utility ac source, a variable-voltage and variable-current dc output is obtained through two basic methods by using static power converters. The first method uses a controllable rectifier to convert the ac source voltage directly into a variable dc voltage in one single stage of power conversion, using phase-controlled converters. The second method converts the ac source voltage to a fixed dc voltage by a diode bridge rectifier and then converts the fixed dc to variable dc voltage with electronic choppers. The second method involves two-stage power conversion, which is dealt with in Chapter 4; the first method is considered in this chapter.

Thyristor converter: The realization of an ac-to-dc variable-voltage converter by means of the silicon-controlled rectifiers (SCR) known as thyristors is discussed in this subsection. The distinct features of the thyristor are given here, without going into the device physics.

Thyristors are four-element (PNPN), three-junction devices with the terminals of anode (A), cathode (K), and gate (G). A gate current is injected by applying a positive voltage between gate and cathode for turning on the device. The device turns on only if the anode is positive compared to cathode at least by 1 V. After turn-on, the device drop is around 1 V for most of the devices. After turning on, the device acts like a diode. Therefore, to turn off the thyristor, the device has to be reverse biased by making the anode negative with respect to the cathode. This is easily achieved with ac input voltage during its negative half-cycle. Turning off the device goes by the name of *device commutation*. The thyristor, unlike the diode, can hold off conduction even when its anode is positive compared to its cathode, by not triggering the gate.

These features make the thyristors ideal devices for ac-to-variable-dc conversion. Instead of diodes in the diode bridge rectifier, thyristors can be substituted, and, by delaying the conduction from their zero crossings, a part of the ac voltage is rectified for feeding to a load. The dc load volt-sec is reduced from the maximum available volt-sec in the half-cycle ac voltage waveform, which in turn varies the average dc load voltage. The line voltage commutates the devices when it reverses polarity each half cycle and applies a negative voltage across the cathode and anode. This method of commutation is known as *natural or line commutation*, and converters using this method are termed *line-commutated* converters.

3.3 PHASE-CONTROLLED CONVERTERS

3.3.1 Single-Phase Controlled Converter

A single-phase controlled-bridge converter is shown in Figure 3.6 with its input and output voltage and current waveforms. The load consists of a resistance and an inductance, and the current is assumed to be continuous and constant. The difference between the diode bridge and this thyristor bridge is that conduction can be delayed in the latter beyond positive zero crossing. The delay is introduced in the form of triggering signals to the gates of the thyristors. The delay angle is measured

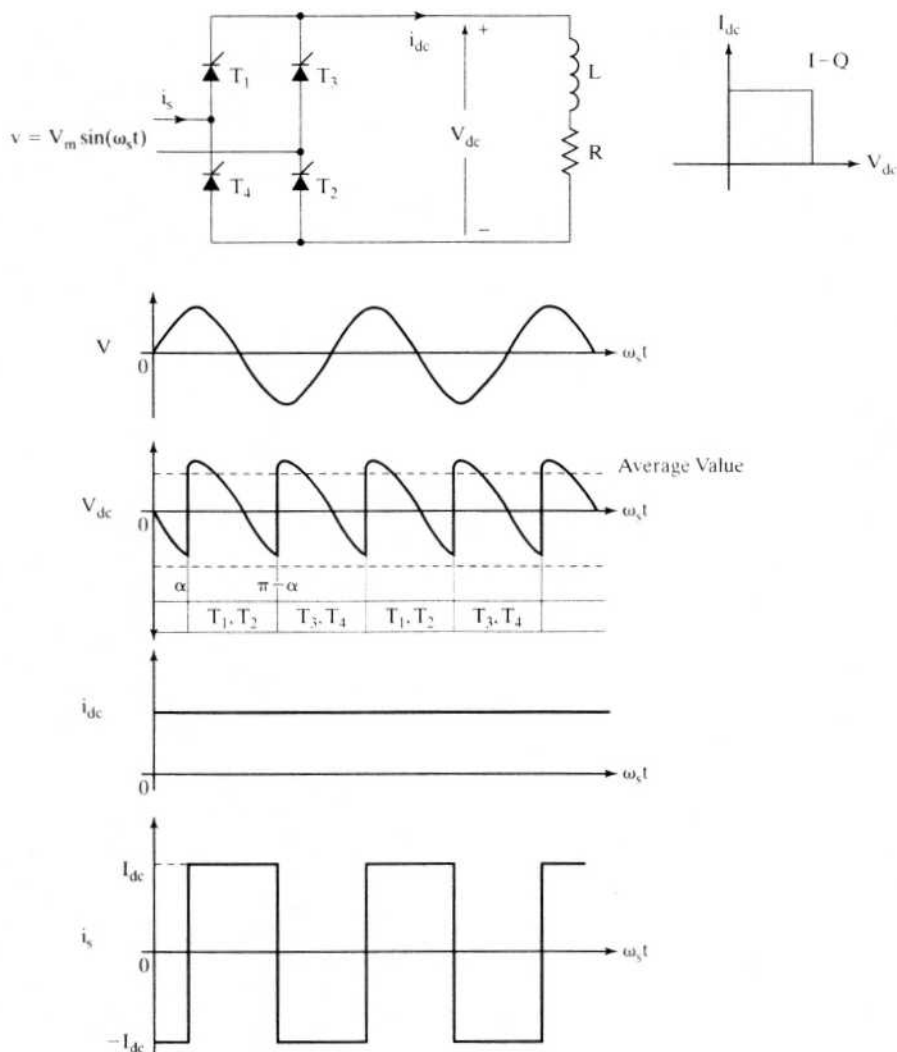


Figure 3.6 A single-phase controlled-bridge converter operating as a rectifier in steady state

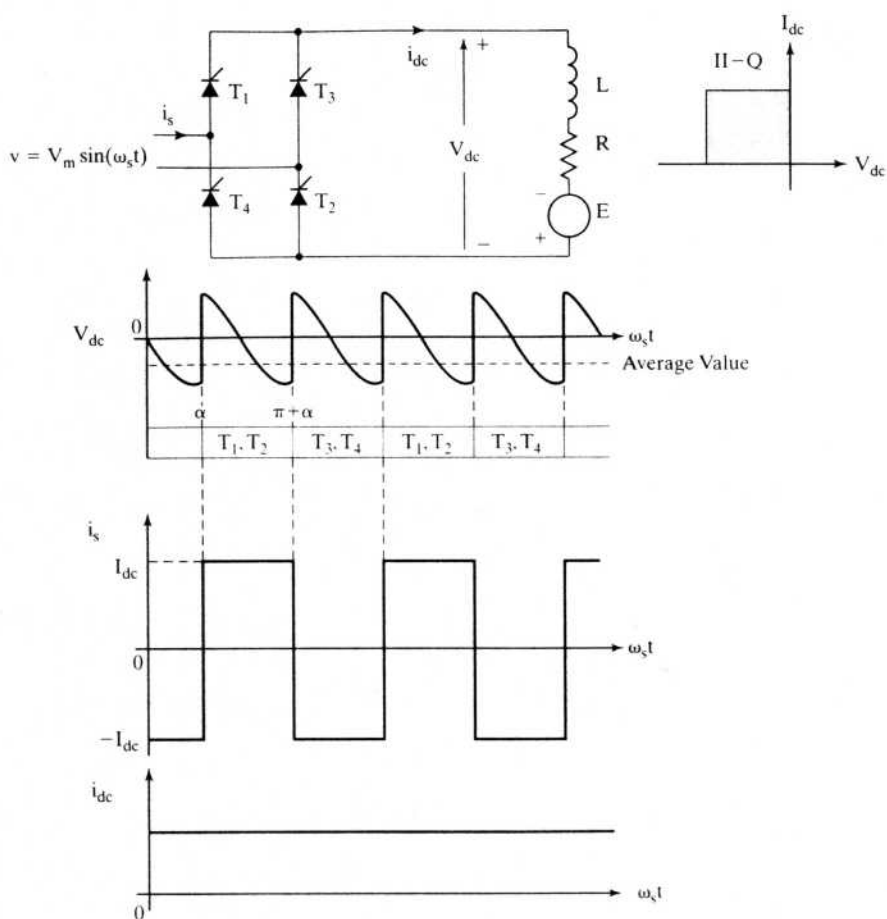


Figure 3.7 Inversion in the steady state of controlled-converter operation

from the zero crossing of the voltage waveform and is generally termed α (in radians) throughout this text. Although the load voltage has both positive and negative volt-seconds, its average is a net positive and is indicated by the horizontal dashed lines denoted as V_{dc} . Assuming continuous load current, this voltage is quantified as

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\alpha + \pi} V_m \sin(\omega_s t) d(\omega_s t) = \frac{2V_m}{\pi} \cos \alpha \quad (3.18)$$

where V_m is the crest of the input ac voltage. Increasing the delay angle to greater than 90° produces a negative voltage on average, as is shown in Figure 3.7. Note that the current is in the same direction, thus contributing to a negative power input. Such a feature is known as inversion. It is tacitly assumed here that there is an emf source in the load contributing to the power transfer from the load to source. Such a load is known as an active load.

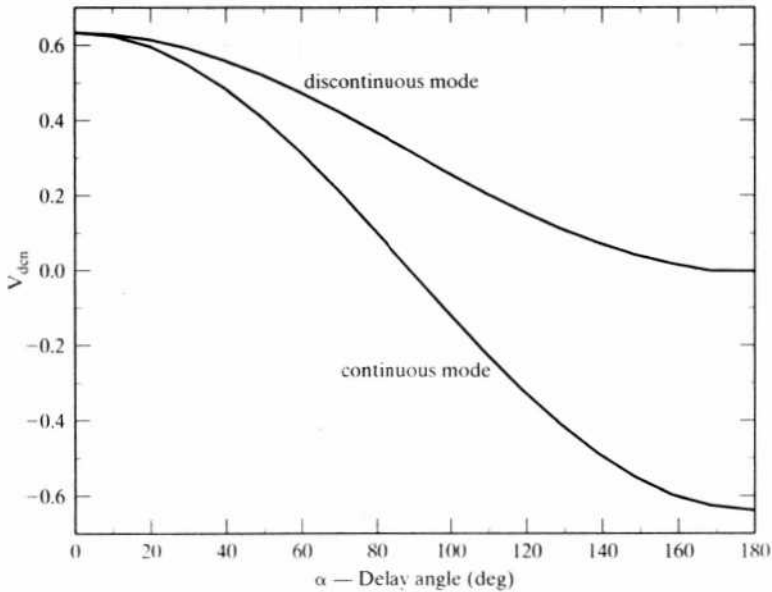


Figure 3.8 Transfer characteristics of a single-phase controlled-bridge rectifier

The load current can be discontinuous; in that case, the average output voltage is derived as

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\alpha+\gamma} V_m \sin(\omega_s t) d(\omega_s t) = \frac{V_m}{\pi} [\cos(\alpha) - \cos(\alpha + \gamma)], \quad V \quad (3.19)$$

where γ is the current conduction angle.

Comparing equations (3.18) and (3.19), it is seen that, for certain values of γ , the output voltage for discontinuous conduction can be greater than that for continuous conduction. For example, let

$$\begin{aligned} \alpha + \gamma &= \pi \\ \alpha &= 30^\circ \end{aligned}$$

Hence, the average output voltage for discontinuous current conduction is

$$V_{dc}(\text{dis}) = \frac{V_m}{\pi} [1.866] \quad (3.20)$$

and, for the same triggering angle, the average output voltage for continuous-current conduction is given by

$$V_{dc}(\text{cont}) = \frac{2V_m}{\pi} \times \frac{1.732}{2} = \frac{1.732V_m}{\pi} \quad (3.21)$$

The transfer characteristics for continuous conduction for an active load and discontinuous conduction for a resistive load are shown in Figure 3.8.

The source inductance delays the current transfer from one pair of conducting thyristors to another set. During this time, the source is short-circuited through the

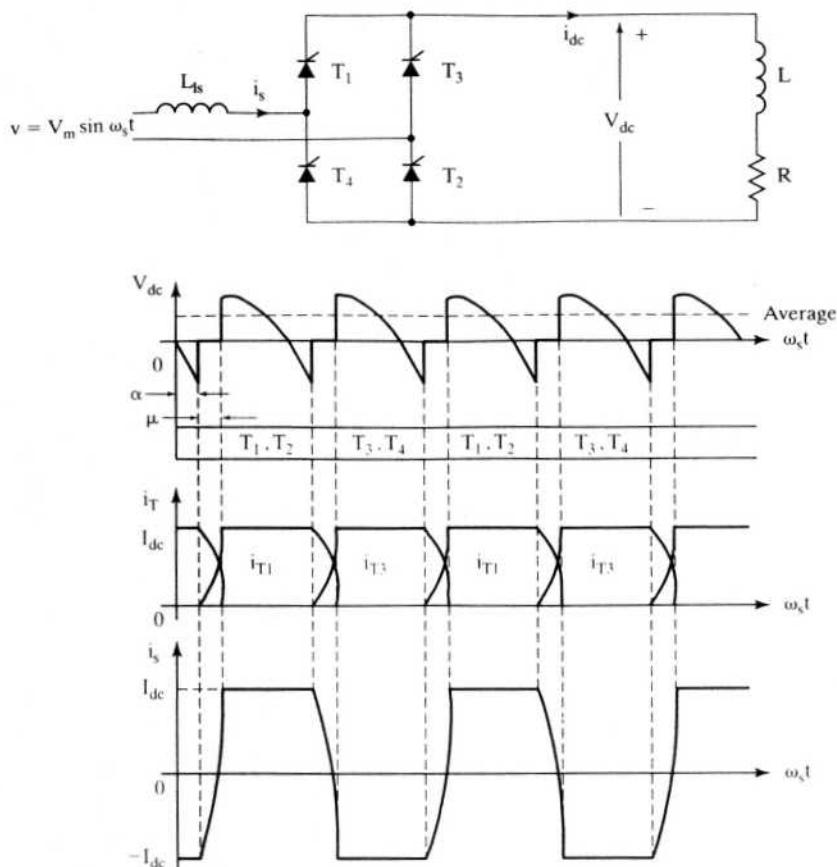


Figure 3.9 The effect of source impedance on the operation of a single-phase controlled rectifier in steady state

source impedance, invariably reducing the load voltage to zero. Hence, the overall effect of source inductance is to reduce the available dc output voltage. Figure 3.9 contains the operational waveforms with source inductance. The source inductance can be introduced by the isolation transformer or by intentionally placed reactors, to reduce the rate of rise of currents in the thyristors. If the source inductance is L_s , the voltage lost due to it is

$$V_s = \frac{1}{\pi} \int_{\alpha}^{\alpha+\mu} V_m \sin(\omega_s t) d(\omega_s t) = \frac{V_m}{\pi} [\cos \alpha - \cos(\alpha + \mu)] \quad (3.22)$$

where μ is the overlap conduction period.

By equating this voltage to the voltage drop in the source reactance, the overlap angle μ is obtained as

$$\mu = \cos^{-1} \left[\cos \alpha - \frac{\pi \omega_s L_s I_{dc}}{V_m} \right] - \alpha \quad (3.23)$$

where I_{dc} is the load current in steady state.

These characteristics are modified when the load includes a counter-emf, as in the case of a dc machine. There is an additional feature in discontinuous operation, with the induced emf appearing across the output of the converter during zero-current intervals, but, if the source emf is instantaneously greater than the back emf, then the conduction starts but will not end immediately when the source emf becomes less than the induced emf, because of the energy stored in the machine inductance and in the external inductance connected in series to the armature of the machine. Therefore, conduction will be prolonged until the energy in these inductances is depleted.

The three-phase controlled full-bridge converter is similar in operation to the single-phase controlled rectifier. Three-phase converters are widely used for both dc and ac motor control. The emphasis is placed in this text on the three-phase converter-controlled dc motor drive. The following section contains the principle of operation, design features of the control circuit, and the characteristics of this converter.

3.3.2 Three-Phase Controlled Converter

A three-phase thyristor-controlled converter is shown in Figure 3.10, and its voltage and current waveforms in the rectifier mode of operation are shown in Figure 3.11. The current is assumed to be continuous for the present. At a given instant, two thyristors are conducting. Assuming that the voltage between phases a and b is maximum, then the thyristors T_1 and T_6 are conducting. The next line voltage to get more positive than ab is ac . At that time, the triggering signal for T_6 will be disabled and that of T_2 will be enabled. Note that the anode of T_2 is more negative than the cathode of T_6 , because line voltage ac is greater than the line voltage ab . That will turn off T_6 and transfer the current from T_6 to T_2 . The delay in current transfer from T_6 to T_2 is dependent on the source inductance. During this current transfer, T_1 , T_6 , and T_2 are all conducting, and the load voltage is the average of the line voltages ab and ac . This phenomenon is the commutation overlap, which results in a reduction in the load voltage. The load current will remain the same during commutation of T_6 . The current in T_6 declines by the same proportion as current in T_2 rises. It is to be

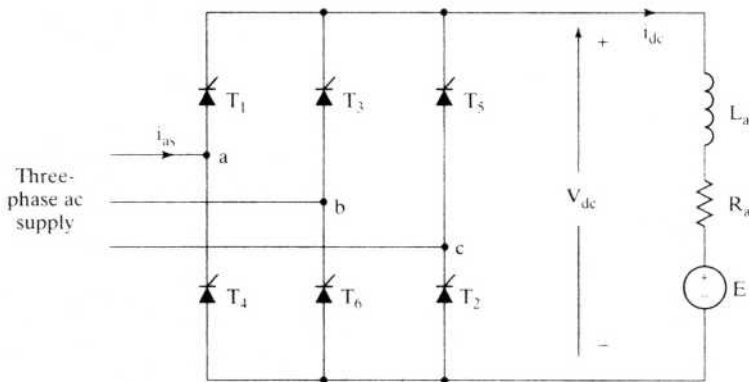


Figure 3.10 Three-phase controlled converter

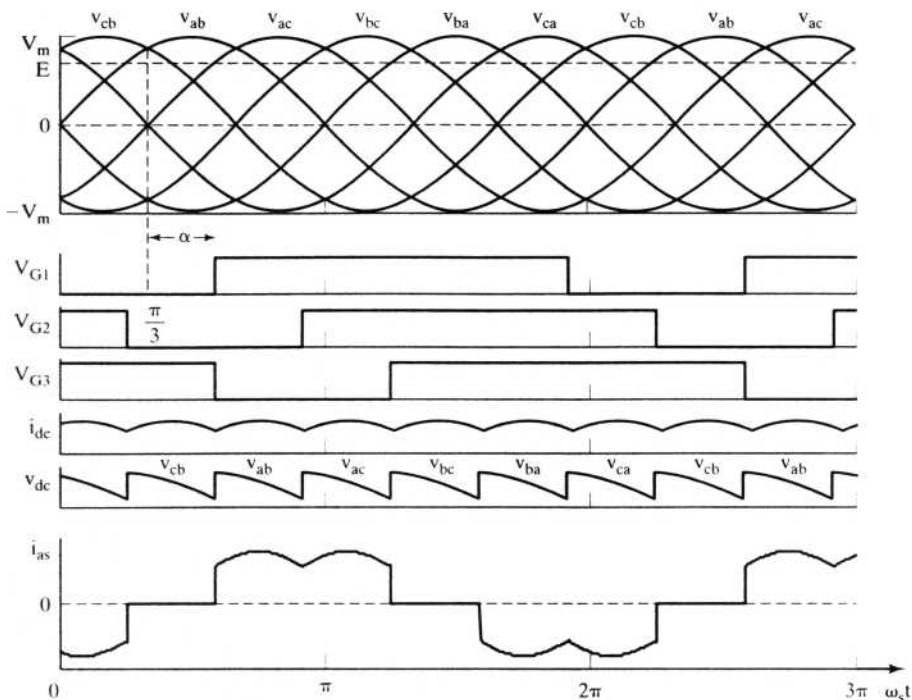


Figure 3.11 Rectification in the three-phase converter in steady state (first-quadrant operation)

observed that the current transfer is effected by the source voltages: voltage ac becoming greater than the voltage ab , resulting in the reverse biasing of T_6 and forward biasing of T_2 . Similarly, it could be seen that the firing/gating sequence is $T_1T_2T_3T_4T_5T_6$ and so on. Also, each of these gating signals is spaced by sixty electrical degrees. The thyristors require small reactors in series to limit the rate of rise of currents and snubbers, which are resistors in series with capacitors across the devices, to limit the rate of rise of voltages when the devices are commutated.

A typical inversion mode of operation is shown in Figure 3.12. Note that this corresponds to a second-quadrant operation of the dc motor drive. The transfer characteristic of the three-phase controlled rectifier is derived as

$$V_{dc} = \frac{1}{\pi/3} \int_{\frac{\pi}{3} + \alpha}^{\frac{2\pi}{3} + \alpha} V_{ab} d(\omega_s t) = \frac{3}{\pi} \int_{\frac{\pi}{3} + \alpha}^{\frac{2\pi}{3} + \alpha} V_m \sin(\omega_s t) d(\omega_s t) = \frac{3}{\pi} V_m \cos \alpha \quad (3.24)$$

The transfer characteristics are very similar to those of the single-phase converter in both the continuous and the discontinuous mode of conduction. The transfer characteristic for the continuous mode of conduction is shown in Figure 3.13. The characteristic is nonlinear. Hence, the use of this converter as a component in a feedback-control system will cause an oscillatory response. This can be explained as follows.

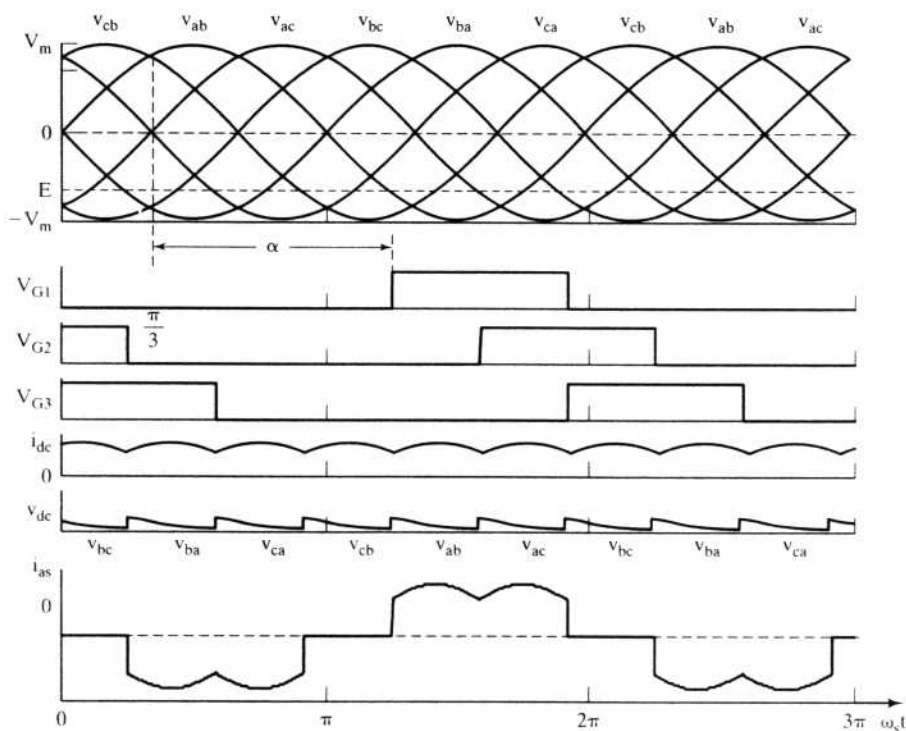


Figure 3.12 Inversion in the three-phase converter in steady state (second-quadrant operation)

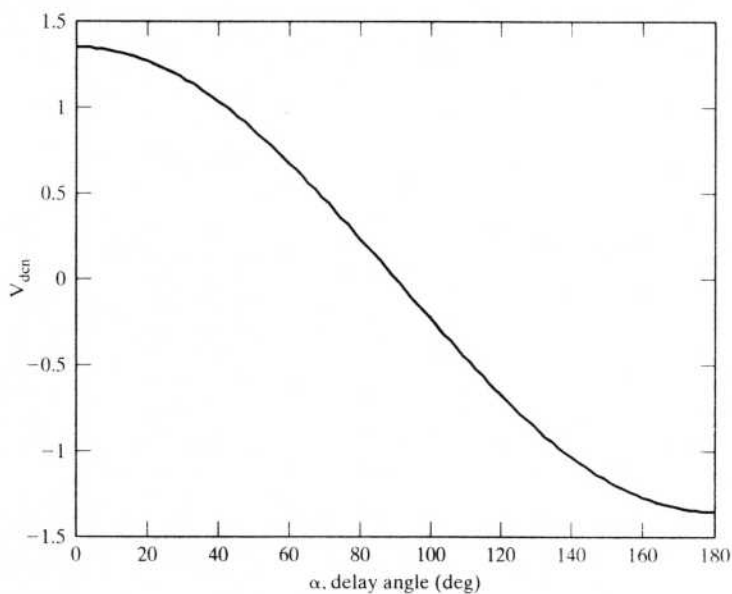


Figure 3.13 Transfer characteristics of a three-phase thyristor-controlled converter

The delay angle will be made a function of speed, current, or position error in a motor-drive system. The error variable is expected to increase or decrease the dc output voltage proportionally. The gain of the converter to a delay angle is not a constant, so it will either overreach or fall short of the required output voltage. This necessitates one more correction in the error signal, causing both time delay and oscillatory response. Such oscillatory responses have been known to create ripple instability in converters.

A control technique to overcome this nonlinear characteristic and the accompanying undesirable dynamic behavior is given in the following. The control input to determine the delay angle is modified to be

$$\alpha = \cos^{-1} \left(\frac{v_c}{V_{cm}} \right) = \cos^{-1} (v_{cn}) \quad (3.25)$$

where v_c is the control input and V_{cm} is the maximum of the absolute value of the control voltage.

Then the dc output voltage is

$$V_{dc} = \frac{3}{\pi} V_m \cos \alpha = \frac{3}{\pi} V_m \cos (\cos^{-1} v_{cn}) = \left[\frac{3}{\pi} V_m \right] v_{cn} = \left[\frac{3}{\pi} \frac{V_m}{V_{cm}} \right] v_c = K_r v_c \quad (3.26)$$

where v_{cn} is the normalized control voltage and K_r is the gain of the converter, defined as

$$K_r = \frac{3}{\pi} \frac{V_m}{V_{cm}} = \frac{3\sqrt{2}V}{\pi V_{cm}} = 1.35 \frac{V}{V_{cm}} \quad (3.27)$$

where V is the rms line-to-line voltage.

Then the modified transfer characteristic is linear with a slope of K_r . The control voltage is normalized to keep its magnitude less than or equal to 1, to be able to obtain the inverse cosine of it.

3.3.3 Control Circuit

The control circuit for the three-phase thyristor converter can be realized in many ways. A schematic of a generic implementation is shown in Figure 3.14. The synchronizing signal is obtained from the line voltage between a and c . The positive-zero crossing of this line voltage forms the starting point for the controller design. The synchronizing signal is multiplexed six times, to have equidistant pulses at 60° intervals. These are decoded to correspond to each gate drive.

The delay angle is obtained from the normalized control voltage v_{cn} through a function generator, so as to make the overall gain of the thyristor converter constant. The delay is incorporated into the synchronized control signal and amplified and fed to the gates of the thyristors. Hence, the maximum limit on the delay angle has to be externally set or commanded. It is essential that sufficient time be given for the thyristor to recover its forward blocking capability. Otherwise, a short of the load and source will occur. Therefore, the maximum limit for the delay angle is usually set in the range of from 150 to 155 degrees. Many adaptive control schemes

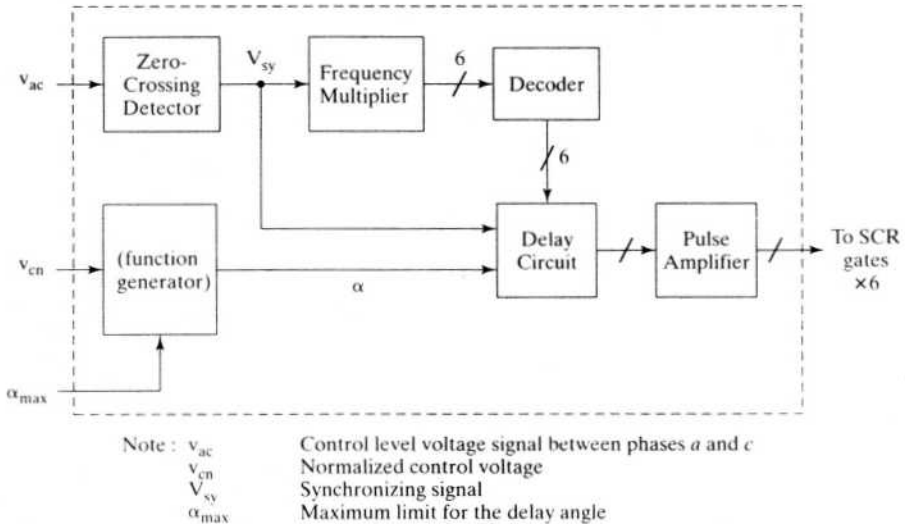


Figure 3.14 Controller schematic for the three-phase converter

adjust this maximum triggering angle as a function of load current. The feature to handle the identification of phase sequence in the ac supply has to be built in, so that the reference synchronization voltage v_{ac} matches with the control circuit.

3.3.4 Control Modeling of the Three-Phase Converter

The converter can be considered as a black box with a certain gain and phase delay for modeling and use in control studies. The gain of the linearized controller-based converter for a maximum control voltage V_{cm} is given in equation (3.27) as

$$K_r = \frac{1.35V}{V_{cm}} \cdot V/V \quad (3.28)$$

The converter is a sampled-data system. The sampling interval gives an indication of its time delay. Once a thyristor is switched on, its triggering angle cannot be changed. The new triggering delay can be implemented with the succeeding thyristor gating. In the meanwhile, the delay angle can be corrected and will be ready for implementation within 60° , i.e., the angle between two thyristors' gating. Statistically, the delay may be treated as one half of this interval; in time, it is equal to

$$T_r = \frac{60/2}{360} \times (\text{time period of one cycle}) = \frac{1}{12} \times \frac{1}{f_s}, s \quad (3.29)$$

For a 60-Hz supply-voltage source, note that the time delay is equal to 1.388 ms. The converter is then modeled with its gain and time delay as

$$G_r(s) = K_r e^{-T_r s} \quad (3.30)$$

and equation (3.30) can also be approximated as a first-order time lag and given as

$$G_r(s) = \frac{K_r}{(1 + sT_r)} \tag{3.31}$$

For most of the drive-system applications, the model given in equation (3.31) is adequate for phase-controlled converters.

Many low-performance systems have a simple controller with no linearization of its transfer characteristic. The transfer characteristic in such a case is nonlinear. Then, the gain of the converter is obtained as a small-signal gain given by

$$K_r = \frac{\delta V_{dc}}{\delta \alpha} = \frac{\delta}{\delta \alpha} \{ 1.35 \text{ V } \cos \alpha \} = -1.35 \text{ V } \sin \alpha \tag{3.32}$$

The gain is dependent on the operating delay angle denoted by α_0 . The converter delay is modeled as an exponential function in Laplace operator s or a first-order lag, describing the transfer function of the converter as in equation (3.31).

3.3.5 Current Source

The key to the control of the machine is to control precisely the electromagnetic torque. This control is achieved in the separately-excited dc machine by controlling its armature current, but the phase-controlled converter provides only a variable voltage output. To make it a controllable current source, a closed-loop control of the dc link current, which in this case is the armature current, is resorted to. A current source can be realized with the phase-controlled converter by incorporating a current feedback loop, as shown in Figure 3.15. Consider a resistive and inductive load combination. The reference current is enforced on the load by comparing it with the actual current in the load. The error current is amplified by a proportional-plus-integral controller and its output is limited so that the control signal will be constrained to be within the maximum triggering angle, α_{max} . The control signal is processed to correspond to linearized operation by the inverse cosine function, and the signals are processed through gate power amplifiers and the converter. Assume that the current reference is a step function and the converter is at rest to start with. The current error will be maximum, which would correspond to minimum triggering angle, thus providing a large voltage across the dc link, i.e., load. This will build up the current in the load, and, when it exceeds the reference value, the current error will reduce from maximum positive to zero. This will enable the triggering angle to be close to $\pi/2$ in this present case, where the dc link voltage, v_{dc} , is not allowed to go below zero. The dc link is constrained to be positive, because the passive load cannot provide for regeneration, and only an active source such as a dc motor will provide the induced emf with the appropriate polarity so that energy can be transferred from the load to the source. The dynamic response of this system with a high proportional gain is shown in Figure 3.16 in normalized units.

The maximum control voltage is 0.7 V, and the control voltage is normalized with respect to this for realizing a linearized controller. The step command of the reference current produces a maximum control voltage, and, correspondingly, the triggering angle is driven to zero to produce the maximum voltage across the dc link. The voltage across the resistor, v_R , starts to increase and that across the induc-

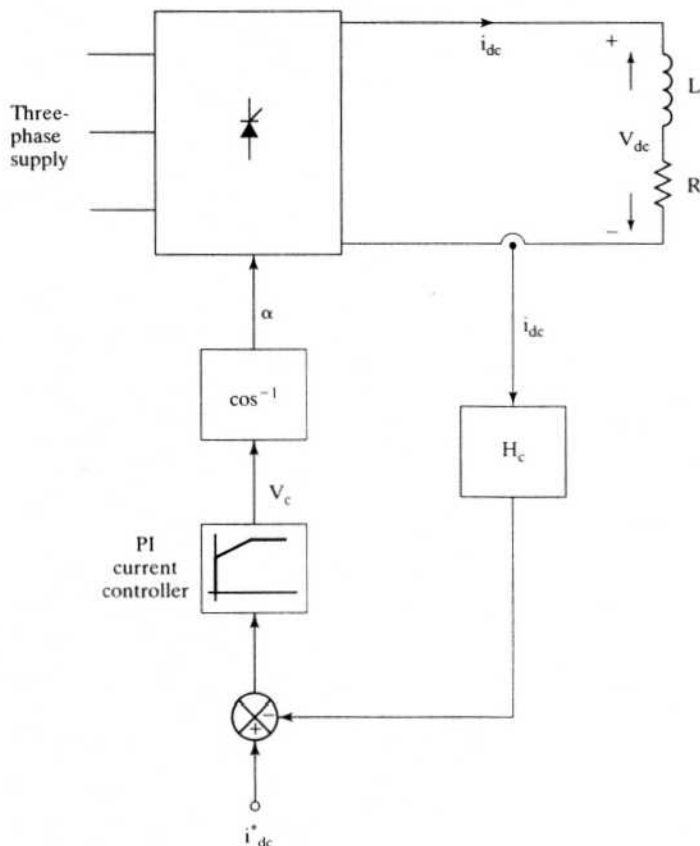


Figure 3.15 Current source with three-phase controlled converter

tance, v_L , begins to decrease. When the current exceeds the reference, the control voltage goes to zero and the triggering angle reaches 90° , to provide an average voltage of zero across the dc link. The current is maintained around the reference value, on average, with a dither, and that dithering is a function of the current controller gains and the load time constant. When the reference current goes to zero, the control voltage becomes zero, and the triggering angle goes to 90° permanently. That forces the current to decay to zero, and the entire voltage of the dc link is borne by the inductance, as shown in Figure 3.16. This type of current source is realized in the 100,000-A range at low voltages for electrolysis in metal processing plants and in a few-thousand-A range at voltages of 600 V to 4,000 V for dc and ac motor drives. The design of the current controller is treated later in this chapter.

3.3.6 Half-Controlled Converter

The converter under study hitherto is a fully controlled bridge converter. Low-power applications can make do with a half-controlled converter, shown in Figure 3.17. The lower half of the bridge has diodes in place of thyristors, thus reducing the cost of

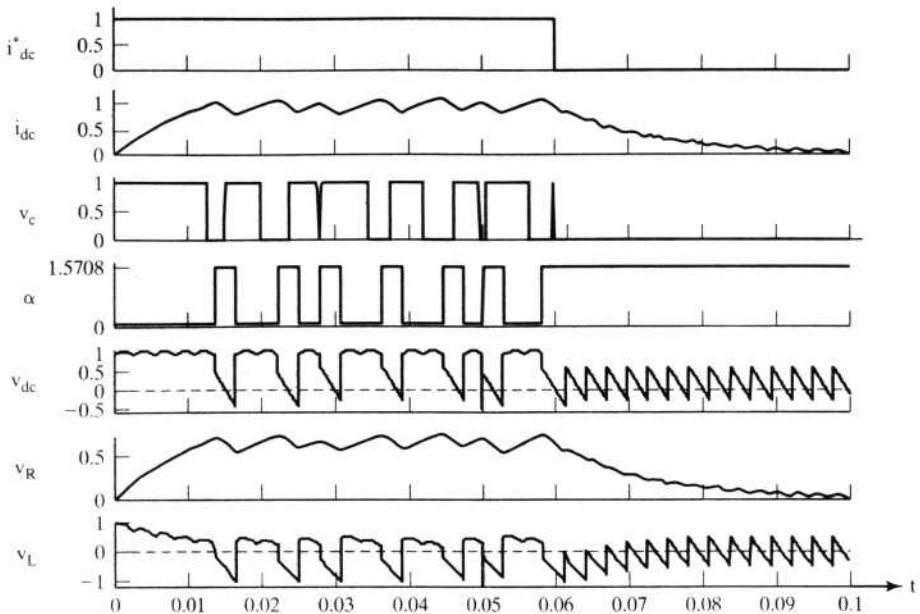


Figure 3.16 Current source operation with R-L load in normalized units

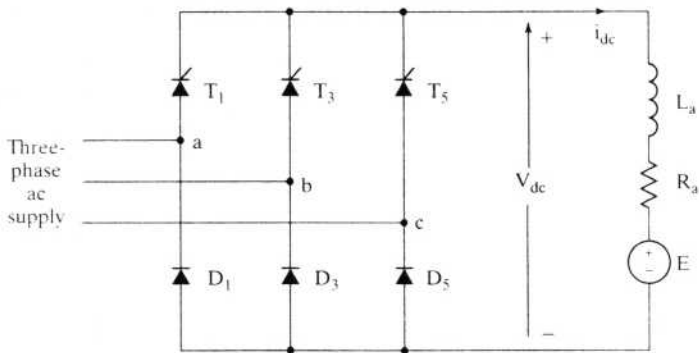


Figure 3.17 Half-controlled three-phase converter

the converter. The control circuit is simplified also. Such a configuration is also possible with a single-phase converter. The half-controlled converters are employed up to 125 hp rating in practice. Note that this converter has only first-quadrant operational capability.

3.3.7 Converters with Freewheeling

A diode is connected across the load as shown in Figure 3.18. The reversal of voltage is not possible now, and hence the converter operates only in the first quadrant, delivering a positive voltage and current. Even though the converter is limited in its capability, the current conduction interval is prolonged by the energy stored in the

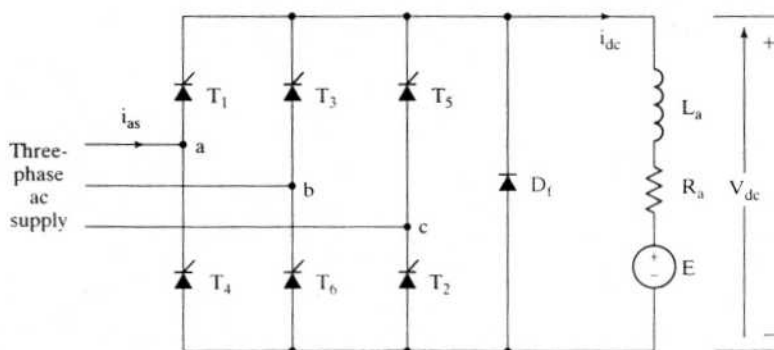


Figure 3.18 Three-phase controlled converter with freewheeling

load inductance. The current continuity has the positive effects of reducing the current ripples in the motor and hence the torque ripples. The waveforms of this freewheeling circuit are shown in Figure 3.19.

There is an alternative way to obtain freewheeling without using a diode across the load. The thyristors of a phase leg can be triggered together and hence can short the load. This, in turn, requires modification of the control circuitry. The configuration utilizing the thyristors has the advantage of minimizing the cost of the converter and making optimal use of the power devices.

3.3.8 Converter Configuration for a Four-Quadrant DC Motor Drive

So far, the converters considered possess one- or two-quadrant operational capability. Figure 3.20 shows the converter configuration for a four-quadrant dc motor drive. It consists of dual three-phase controlled-bridge converters in parallel, with their output polarities reversed. Converter 1 gives both positive and negative dc output voltages with a positive current output, catering to the first- and second-quadrant operation of the dc motor drive, respectively. This converter is hereafter referred to as the forward converter. Similarly, converter 2 delivers negative current to the motor with positive and negative dc output voltages. Such operation encompasses the fourth- and third-quadrant performance of the dc motor drive, respectively. Because of it, this converter is hereafter referred to as a reverse converter. Transitioning from forward to reverse operation in the motor and corresponding changes in the converters require special logic control circuits to avoid short-circuiting the ac supply and for safe operation. The phase-controlled converter has two quadrants of operation: quadrants I and II, or quadrants III and IV, depending on the connection of the converter relative to the machine armature. Note that quadrants I and IV or quadrants II and III are more useful in many unidirectional motion control applications than the ones provided by the converter. That will enable both the motoring and generating/braking torque production contributing to faster acceleration and deceleration in one direction of rotation. The control strategy for a four-quadrant dc motor drive is discussed later in this chapter.

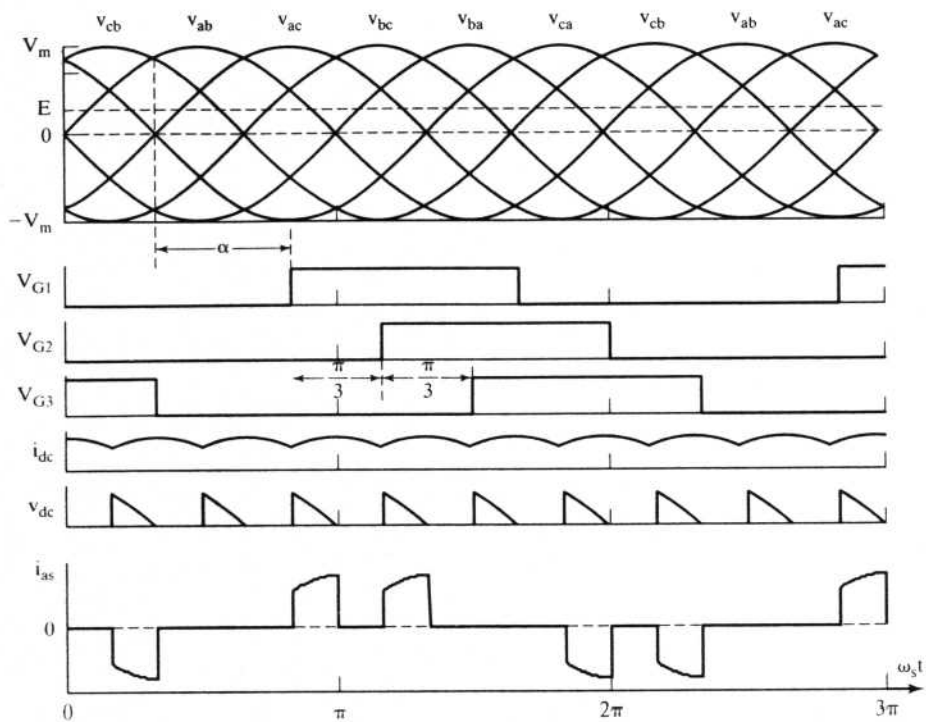


Figure 3.19 Three-phase controlled converter in freewheeling operation

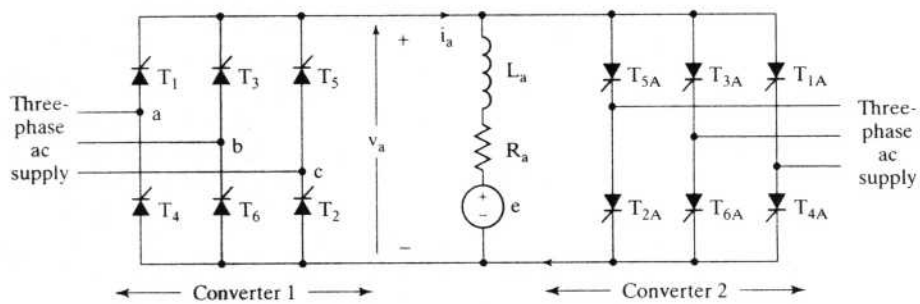


Figure 3.20 Dual three-phase thyristor converter for four-quadrant drive

3.4 STEADY-STATE ANALYSIS OF THE THREE-PHASE CONVERTER-CONTROLLED DC MOTOR DRIVE

3.4.1 Average Analysis

A separately-excited dc motor is fed from a three-phase converter and is operated in one rotational direction, say, in the first and fourth quadrants of torque–speed characteristics. The steady-state performance of this motor drive is described in this section. The steady-state performance, when combined with the load characteristics,

provides the basis for evaluating the suitability of the motor drive for the given application. The steady-state performance is developed by assuming that the average values only are considered. Indirectly, it is implied that the average current produces an average torque, which, in combination with load torque, determines the average speed. In that process, the quasi-transients are neglected.

Then the armature voltage equation for the motor in steady state is

$$v_a = R_a i_a + e \quad (3.33)$$

and, in terms of average values (by denoting the variables in capital letters or with a subscript 'av'),

$$V_a = R_a I_a + K\Phi_f \omega_{\text{mav}} \quad (3.34)$$

Average electromagnetic torque is given by

$$T_{\text{av}} = K\Phi_f I_a \quad (3.35)$$

and, from a previous derivation, the average dc link voltage is

$$V_a = 1.35V \cos \alpha \quad (3.36)$$

where V is the rms line-to-line ac voltage in a three-phase system.

Then the electromagnetic torque is expressed in terms of delay angle and speed, from equations (3.34) to (3.36), as

$$T_{\text{av}} = K\Phi_f \left\{ \frac{V_a - K\Phi_f \omega_{\text{mav}}}{R_a} \right\} = K\Phi_f \left\{ \frac{1.35V \cos \alpha - K\Phi_f \omega_{\text{mav}}}{R_a} \right\} \quad (3.37)$$

The equation (3.37) is normalized by dividing the average torque by the rated torque:

$$T_{\text{en}} = \frac{T_{\text{av}}}{T_{\text{cr}}} = \frac{T_{\text{av}}}{K\Phi_{\text{fr}} I_{\text{ar}}} = \frac{K\Phi_f \{1.35V \cos \alpha - K\Phi_f \omega_{\text{mav}}\}}{K\Phi_{\text{fr}} I_{\text{ar}} R_a} = \frac{[1.35V \cos \alpha - K\Phi_f \omega_{\text{mav}}]}{I_{\text{ar}} R_a} \Phi_{\text{fn}} \quad (3.38)$$

Dividing the numerator and denominator of equation (3.38) by the rated motor voltage, V_r , leads to

$$T_{\text{en}} = \frac{\left[1.35 \frac{V}{V_r} \cos \alpha - \frac{K\Phi_f \omega_{\text{mav}}}{V_r} \right]}{\frac{I_{\text{ar}} R_a}{V_r}} \Phi_{\text{fn}} \quad (3.39)$$

and, noting that

$$V_r = K\Phi_{\text{fr}} \omega_{\text{mr}} \quad (3.40)$$

$$R_{\text{an}} = \frac{R_a I_{\text{ar}}}{V_r}, \text{ p.u.} \quad (3.41)$$

the normalized electromagnetic torque is given by

$$T_{\text{en}} = \frac{[1.35V_n \cos \alpha - \Phi_{\text{fn}} \omega_{\text{mn}}]}{R_{\text{an}}} \Phi_{\text{fn}}, \text{ p.u.} \quad (3.42)$$

where

$$\Phi_{fn} = \frac{\Phi_f}{\Phi_{fr}}, \text{ p.u.} \quad (3.43)$$

$$\omega_{mn} = \frac{\omega_{mav}}{\omega_{mr}}, \text{ p.u.} \quad (3.44)$$

and

$$V_n = \frac{V}{V_r}, \text{ p.u.} \quad (3.45)$$

The normalized equation (3.42) deserves careful scrutiny for use in steady-state performance computation. Positive or motoring torque is produced when the numerator of (3.42) is positive, i.e.,

$$1.35V_n \cos \alpha - \Phi_{fn}\omega_{mn} > 0 \quad (3.46)$$

or

$$\cos \alpha > \frac{\Phi_{fn}\omega_{mn}}{1.35V_n} \quad (3.47)$$

If $\cos \alpha$ is less than the right-hand side of (3.47), then there is no torque generation in the machine. For some positive values of α , the numerator can become negative, but that will not produce regeneration, since there will be no current flow from the machine to the source with only one converter. The induced emf of the machine will be greater than the applied voltage, thus blocking the conduction of thyristors. If an antiparallel converter is available and is capable of conducting current in the reverse direction to the motoring operation, then regeneration is achieved by decreasing the applied voltage compared with the value of the induced emf. That enables the machine to generate current from the difference between its induced emf and the applied voltage. This step results in power flow from the machine to the ac source.

The electromagnetic torque equation (3.42) expressed as a function of normalized flux, speed, and triggering angle delay has design use. The controller for the converter requires a certain precision in its triggering angle delay. Its resolution can be found by, for instance, finding the minimum and maximum triggering delay angles from the torque requirements at the minimum and maximum speeds of operation by using this expression. The range of triggering delay angle and the finesse with which it needs to be controlled is given by the resolution of speed or torque control.

Example 3.3

Consider a motor drive with $R_{an} = 0.1$ p.u., $\Phi_{fn} = 1$ p.u., $V_n = 1.1$ p.u. and extreme load operating points $T_{e1(\min)} = 0.1$ p.u., $\omega_{mn(\min)} = \omega_{mn1} = 0.1$ p.u., $T_{e2(\max)} = 1$ p.u., and $\omega_{mn(\max)} = \omega_{mn2} = 1$ p.u.

- (i) Find the normalized control voltages to meet these operating points.

- (ii) Compute the change in control voltages required for a simultaneous change of $\Delta T_{en} = 0.02$ p.u. and $\Delta \omega_{mn} = 0.01$ p.u. for both the extreme operating points. From this, calculate the resolution required for the control voltage.

Solution Assume that the controller is linearized.

$$\therefore \alpha = \cos^{-1} \left\{ \frac{V_c}{V_{cn}} \right\} = \cos^{-1} \{ V_{cn} \}$$

from which the electromagnetic torque is,

$$T_{en} = \left[\frac{1.35 V_n V_{cn} - \Phi_{fn} \omega_{mn}}{R_{an}} \right] \Phi_{fn}, \text{ p.u.}$$

where V_{cn} is the normalized control voltage for a given steady-state operating point and is obtained as

$$V_{cn} = \frac{T_{en} R_{an} + \Phi_{fn} \omega_{mn}}{1.35 V_n}$$

Since $\Phi_{fn} = 1$ p.u., the control voltage for minimum torque and speed is

$$V_{cn1} = \frac{T_{en1} R_{an} + \Phi_{fn} \omega_{mn1}}{1.35 V_n} = \frac{0.1 * 0.1 + 0.1}{1.35 * 1.1} = 0.074 \text{ p.u.}$$

Similarly for maximum torque and speed, the control voltage is

$$V_{cn2} = \frac{T_{en2} R_{an} + \Phi_{fn} \omega_{mn2}}{1.35 V_n} = \frac{1 * 0.1 + 1}{1.35 * 1.1} = 0.74 \text{ p.u.}$$

Incremental control voltage generates incremental torque and speed as

$$V_{cn} + \delta V_{cn} = \frac{R_{an}(T_{en} + \delta T_{en}) + \omega_{mn} + \delta \omega_{mn}}{1.35 V_n}$$

$$\text{For both changes, } \delta V_{cn} = \frac{R_{an} \Delta T_{en} + \delta \omega_{mn}}{1.35 V_n}$$

Dividing δV_{cn} by V_{cn} gives an expression in terms of steady-state operating points as

$$\frac{\delta V_{cn}}{V_{cn}} = \frac{R_{an} \delta T_{en} + \delta \omega_{mn}}{R_{an} T_{en} + \omega_{mn}}$$

$$\delta T_{en} = 0.02 \text{ p.u., } \delta \omega_{mn} = 0.01 \text{ p.u., } T_{en1} = 0.1 \text{ p.u., } \omega_{mn1} = 0.1 \text{ p.u., } T_{en2} = 1 \text{ p.u., } \omega_{mn2} = 1 \text{ p.u.}$$

$$\text{For } T_{en1}, \omega_{mn1}: \frac{\delta V_{cn}}{V_{cn}} = \frac{0.1 * 0.02 + 0.01}{0.1 * 0.1 + 0.1} = 0.109$$

$$\text{For } T_{en2}, \omega_{mn2}: \frac{\delta V_{cn}}{V_{cn}} = \frac{0.1 * 0.02 + 0.01}{0.1 * 1 + 1} = 0.0109$$

Therefore, the resolution required in control voltage is

$$\delta V_{cn} = 0.109 * V_{cn} = 0.109 * 0.074 = 0.008066 \text{ p.u.}$$

Example 3.4

A separately-excited dc motor has 0.05 p.u. resistance and is fed from a three-phase converter. The normalized voltage and field flux are 1 p.u. Draw the torque–speed characteristics in the first quadrant for constant delay angles of 0, 30, 45, and 60 degrees. Indicate the safe operating region if the maximum torque limit is 2.5 p.u.

Solution

$$T_{en} = \frac{[1.35V_n \cos \alpha - \Phi_{fn}\omega_{mn}]}{R_{an}} \Phi_{fn}, \text{ p.u.}$$

Substituting the given values yields

$$T_{en} = 20[1.35 \cos \alpha - \omega_{mn}], \text{ p.u.}$$

The torque–speed characteristics for various angles of delay are shown in Figure 3.21. The safe operating region is shaded in the figure.

3.4.2 Steady-State Solution Including Harmonics

This section considers the steady-state analysis of the dc motor drive with the actual voltage waveforms and not the average values of the input voltages as in the previous section. The advantage of considering the actual current waveforms is to accu-

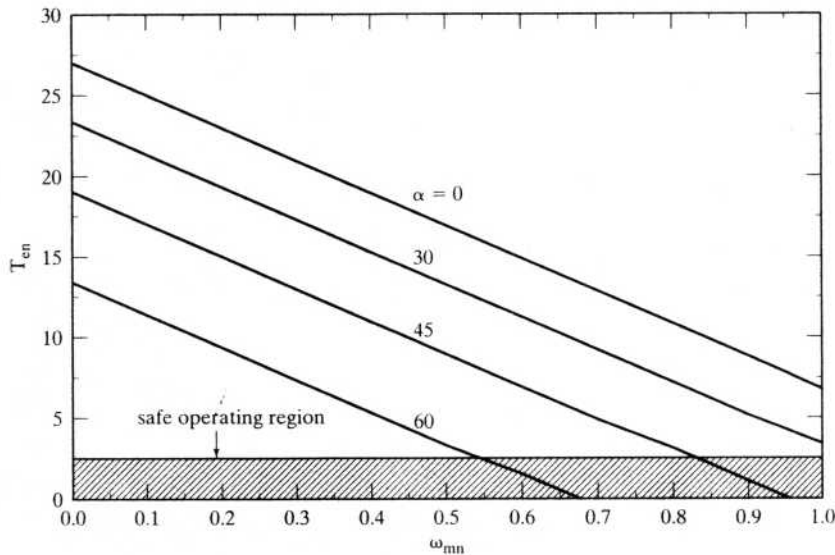


Figure 3.21 Torque–speed characteristics as a function of the triggering delay angle

rately predict its electromagnetic torque and hence the magnitude of the torque pulsations. The information on the torque pulsations is necessary for applications such as position servo drives and antenna drives. Additionally, the actual computation of current waveforms assists in the evaluation of harmonic losses and hence of the heating effects on the machine, which will be instrumental in the derating consideration, as explained in a later section.

For this analysis, the motor is considered to be in steady state, i.e., the speed of the machine and the field current are assumed to be constant. In that case, the equation of the motor is sufficiently provided by its electrical part, given as

$$R_a i_a + L_a \frac{di_a}{dt} + K_b \omega_m = v_a \quad (3.48)$$

where

$$v_a = V_m \sin(\omega_s t + \pi/3 + \alpha), 0 < \omega_s t < \pi/3 \quad (3.49)$$

and for each $\pi/3$ duration, the same is valid. The current may be continuous or discontinuous, depending on the speed, the input line voltage, the triggering angle, and the impedance of the motor. For the sake of simplicity, the commutation effect is neglected in this analysis. It can be easily incorporated for the design calculations without sacrificing the elegance of the present solution. The induced emf is a constant under the assumption of constant speed, and, hence, the solution of the above equation is given by

$$i_a(t) = \left(\frac{V_m}{|Z_a|} \right) \{ \sin(\omega_s t + \pi/3 + \alpha - \beta) - \sin(\pi/3 + \alpha - \beta) e^{-t/T_a} \} - \left(\frac{E}{R_a} \right) (1 - e^{-t/T_a}) + i_{ai} e^{-t/T_a} \quad (3.50)$$

where $\omega_s = 2\pi f_s$, $\beta = \tan^{-1}(\omega_s \cdot L_a/R_a)$ = machine impedance angle, T_a = armature time constant = L_a/R_a , i_{ai} = initial value of current at time $t = 0$, $E = K_b \cdot \omega_m$ = induced emf, V_m = peak value of the line-line input ac voltage, and motor electrical impedance $Z_a = R_a + j\omega_s L_a$.

The initial armature current i_{ai} has to be evaluated to obtain the complete solution, and it could be achieved from one other available piece of information. That is that the armature current repeats itself every 60 electrical degrees, similarly to the input voltage. Such a unique situation gives rise to a boundary condition:

$$i_a(\omega_s t) = i_a(\omega_s t + \pi/3) \quad (3.51)$$

That is, the initial armature current will be equal to the current at the end of 60 electrical degrees. Therefore, evaluating the current at a time corresponding to the 60th electrical degree and equating it to the initial value of the armature current, i_{ai} , as follows, leads to

$$i_a\left(\frac{\pi}{3\omega_s}\right) = i_{ai}$$

$$\begin{aligned} &= \left(\frac{V_m}{|Z_a|} \right) \left\{ \sin \left(\frac{2\pi}{3} + \alpha - \beta \right) - \sin \left(\frac{\pi}{3} + \alpha - \beta \right) e^{-\left(\frac{\pi}{3\omega_s T_s} \right)} \right\} \\ &- \left(\frac{E}{R_a} \right) \left(1 - e^{-\left(\frac{\pi}{3\omega_s T_s} \right)} \right) + i_{ai} e^{-\left(\frac{\pi}{3\omega_s T_s} \right)} \end{aligned} \tag{3.52}$$

Rearranging the initial current on one side, it is evaluated as a function of only machine parameters, input voltage, triggering angle, and speed and given as

$$i_{ai} = \frac{\left(\frac{V_m}{|Z_a|} \right) \left\{ \sin \left(\frac{2\pi}{3} + \alpha - \beta \right) - \sin \left(\frac{\pi}{3} + \alpha - \beta \right) e^{-\left(\frac{\pi}{3\omega_s T_s} \right)} \right\} - \frac{E}{R_a} \left(1 - e^{-\left(\frac{\pi}{3\omega_s T_s} \right)} \right)}{1 - e^{-\left(\frac{\pi}{3\omega_s T_s} \right)}} \tag{3.53}$$

By using this initial value of the current, the armature current for the complete cycle can be evaluated. The present approach, using the boundary matching condition, is a powerful technique for evaluating the steady state directly, without going through the transients in the solution of the differential equation. This technique is widely applied in the evaluation of steady-state performance in variable-speed dc and ac motor drives and is adopted throughout this text. This technique also gives a closed-form solution that could be used in the analysis and design of the drive systems. Figure 3.22 shows the phase converter output voltage, armature current, and induced emf, using the above solution procedure. Although the input armature voltage contains considerable harmonic, the current has much less harmonic, because of the high harmonic impedance of the machine. The useful electromagnetic torque is

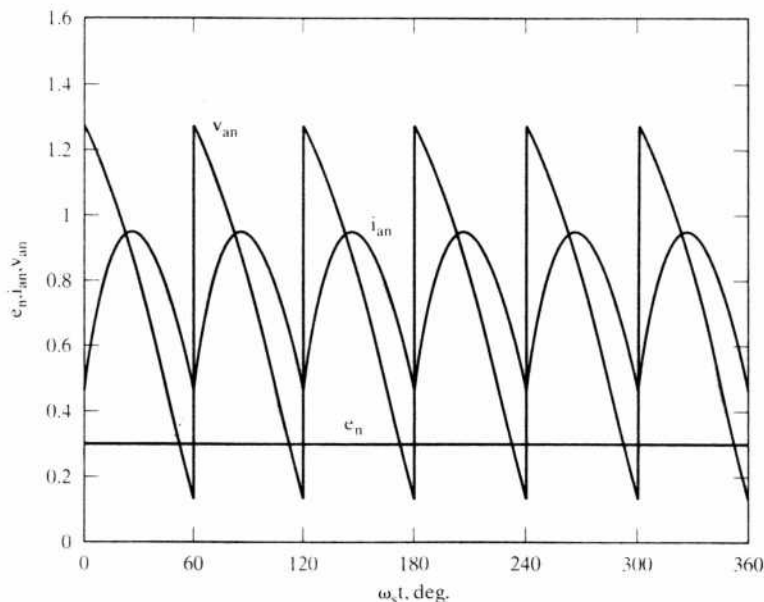


Figure 3.22 The output voltage, the armature current, and the induced emf of the phase converter

the average; the harmonic torques contribute to losses only. Note that the electromagnetic torque is very similar to the armature current in shape, with the field current being maintained constant.

3.4.3 Critical Triggering Angle

The triggering angle corresponding to when the armature current is barely continuous is called the critical triggering angle, α_c . It is evaluated by equating the initial armature current, i_{ai} , to zero in equation (3.53). That is given as

$$\alpha_c = \beta + \cos^{-1} \left\{ \frac{(E/V_m)}{c_1} \cdot \frac{1}{\cos \beta} \cdot (1 - e^{-(\pi/3 \tan \beta)}) \right\} - \frac{\pi}{3} + \theta_1 \quad (3.54)$$

where

$$\left. \begin{aligned} c_1 &= \sqrt{a_1^2 + b_1^2} \\ a_1 &= \frac{\sqrt{3}}{2} \\ b_1 &= \frac{1}{2} - e^{-(\frac{\pi}{3 \tan \beta})} \\ \theta_1 &= \tan^{-1} \left(\frac{b_1}{a_1} \right) \end{aligned} \right\} \quad (3.55)$$

This expression for critical triggering angle is a function of the induced emf, input line-to-line ac voltage, and machine impedance angle, β . The induced emf and the input ac line voltage can be expressed as a single variable, because the input line-to-line voltage is usually a constant, and that leaves the dependence of the critical triggering angle on two variables, E/V_m and β , only. Figure 3.23 shows the critical triggering angle vs. E/V_m for various values of β . It is to be understood that these curves are not machine-specific and could be used to calculate the critical triggering angles for any dc motor-drive system. An increase in the triggering angle beyond its critical value implies that the armature current will become discontinuous.

When one is computing α_c by using equation (3.54), care has to be exercised in the argument of arccosine. If the argument exceeds one, then either E/V_m or β has to be changed to limit it to one. Further, the arccosine term has to be less than the machine impedance angle, β , to be meaningful; accordingly, the computation has to be terminated when α_c becomes less than or equal to zero.

3.4.4 Discontinuous Current Conduction

When the current becomes discontinuous, the voltage across the machine then is the induced emf itself. The steady state under such a condition can be computed from the following relationship:

$$R_a i_a + L_a \frac{di_a}{dt} + E = V_a, \quad 0 < \omega_s t < \omega_s t_x \quad (3.56)$$

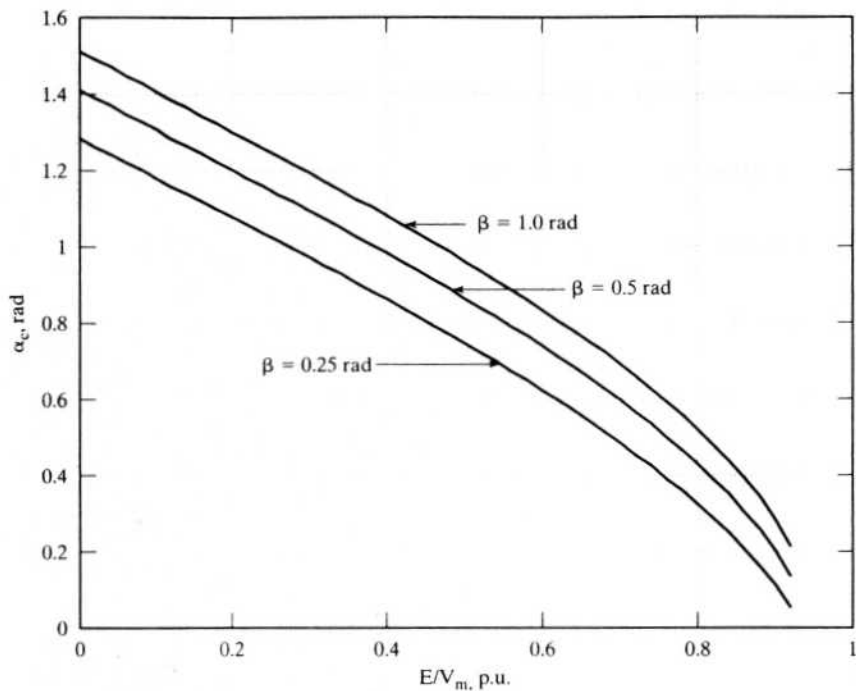


Figure 3.23 The critical triggering angle vs. E/V_m for various values of β

$$i_a = 0, \omega_s t_x < \omega_s t < \pi/3 \quad (3.57)$$

$$v_a = V_m \sin(\omega_s t + \pi/3 + \alpha), 0 < \omega_s t < \omega_s t_x \quad (3.58)$$

$$= E, \omega_s t_x < \omega_s t < \pi/3 \quad (3.59)$$

and $\omega_s t_x = \theta_x$. Time t_x corresponds to the current conduction time, and the waveforms are shown in Figure 3.24. The time t_x can be evaluated from the armature current solution by equating it to zero at time $t = t_x$, as follows:

$$i_a(t_x) = 0 = \frac{V_m}{Z_a} \left[\sin(\omega_s t_x + \pi/3 + \alpha - \beta) - \sin(\pi/3 + \alpha - \beta) e^{-\frac{t_x}{T_i}} \right] - \frac{E}{R_a} \left(1 - e^{-\frac{t_x}{T_i}} \right) \quad (3.60)$$

This equation can be solved iteratively for t_x by using the Newton-Raphson technique. The current conduction time t_x is dependent on the machine parameters, speed (which is contained in the induced emf term), ac input line-to-line voltage, and triggering angle. This triggering angle will be greater than the critical triggering angle evaluated in the earlier section. Because of its dependence on more than two variables, the current conduction time vs. all the variables cannot be contained in a two or three dimensional plot as in the case of the critical triggering angle. It needs to be evaluated for each operating condition. The discontinuous current has a rich content of harmonics compared to the continuous current, resulting in high torque pulsations that might be undesirable for some applications.

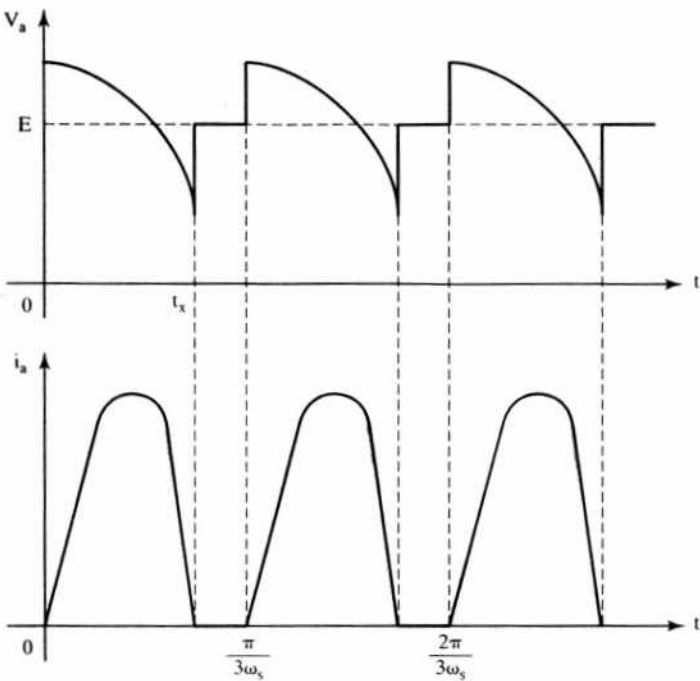


Figure 3.24 Discontinuous current conduction waveforms

Figure 3.25 shows the armature current, the voltage across the armature terminals, and the induced emf for a typical discontinuous conduction. In the discontinuous mode of conduction, note that the linearity of the output voltage with reference voltage will be lost. This has implications in the control contributing to an overall sluggish response of the drive system. The current discontinuity can be overcome by the addition of an external inductor or by suitably designing the dc machine with the required armature inductance to obtain a continuous current. The latter approach is feasible at the system level of planning mostly for new installations. The approach of using an external inductor is the only practical recourse for existing installations; the replacement of the dc machine is an expensive alternative.

Example 3.5

The details and parameters of a separately-excited dc machine are

$$100 \text{ hp}, 500 \text{ V}, 1750 \text{ rpm}, 153.7 \text{ A}, R_a = 0.088 \, \Omega, L_a = 0.00183 \text{ H}, K_b = 2.646 \text{ V}/(\text{rad/sec})$$

The machine is supplied from a three-phase controlled converter whose ac input is from a three-phase 415 V, 60 Hz utility supply. Assume that the machine is operating at 100 rpm with a triggering angle delay of 65°. Find the maximum air gap torque ripple at this operating point.

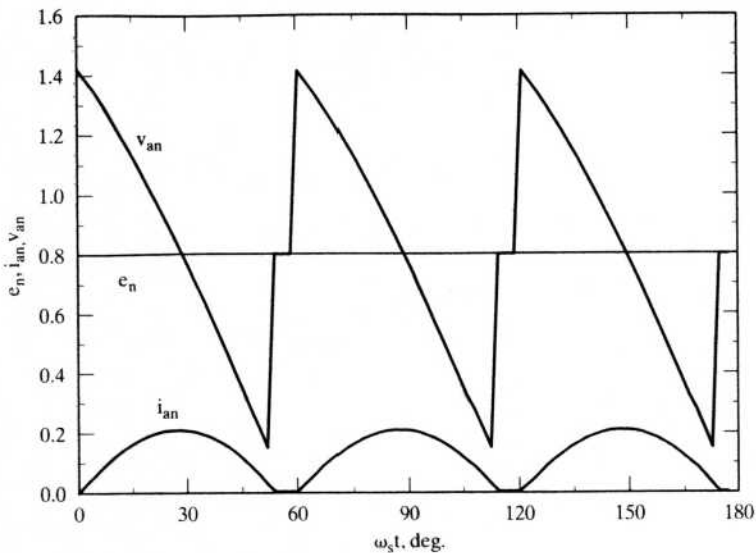


Figure 3.25 The armature current, applied voltage, and induced emf for a typical discontinuous conduction

Solution To find the current ripple, it is essential to determine whether the current is continuous for the given triggering delay by evaluating the critical triggering angle. It is found as follows:

Rotor speed, $\omega_m = \frac{2\pi N_r}{60} = \frac{2\pi * 100}{60} = 10.48 \text{ rad/sec}$

Induced emf, $E = K_b \omega_m = 2.646 * 10.48 = 27.7 \text{ V}$

Peak input voltage, $V_m = \sqrt{2} * 415 = 586.9 \text{ V}$

Input angular frequency, $\omega_s = 2\pi f_s = 2\pi * 60 = 376.99 \text{ rad/sec}$

Armature time constant, $T_a = \frac{L_a}{R_a} = \frac{0.00183}{0.088} = 0.0208 \text{ sec}$

Machine impedance, $Z_a = \sqrt{R_a^2 + \omega_s^2 L_a^2} = 0.6955 \Omega$

Machine impedance angle, $\beta = \tan^{-1}\left(\frac{\omega_s L_a}{R_a}\right) = 1.444 \text{ rad}$

The critical triggering angle is

$$\alpha_c = B + \cos^{-1}\left\{\frac{E/V_m}{c_1} \cdot \frac{1}{\cos \beta} \cdot (1 - e^{-(\pi/3 \tan \beta)})\right\} - \frac{\pi}{3} + \theta_1$$

where

$$a_1 = \frac{\sqrt{3}}{2} = 0.866$$

$$b_1 = \frac{1}{2} - e^{-(\frac{\pi}{3 \tan \beta})} = -0.375$$

$$c_1 = \sqrt{a_1^2 + b_1^2} = 0.9437$$

$$\theta_1 = \tan^{-1}\left(\frac{b_1}{a_1}\right) = -0.4086 \text{ rad}$$

from which the critical angle is obtained as $\alpha_c = 1.5095 \text{ rad} = 86.48^\circ$. The triggering angle α is 65° , which is less than the critical triggering angle; therefore, the armature current is continuous. Having determined that the drive system is in continuous mode of conduction, we use the relevant equations to calculate the initial current, given by

$$i_{a1} = \frac{\left(\frac{V_m}{|Z_a|}\right)\left\{\sin\left(\frac{2\pi}{3} + \alpha - \beta\right) - \sin\left(\frac{\pi}{3} + \alpha - \beta\right)e^{-\left(\frac{\pi}{\omega_r T_s}\right)}\right\} - \frac{E}{R_a}(1 - e^{-\left(\frac{\pi}{\omega_r T_s}\right)})}{1 - e^{-\frac{\pi}{\omega_r T_s}}} = 2308.1 \text{ A}$$

The peak armature current is found by having $\omega_s t = \pi/6$, i.e., at the midpoint of the cycle. This is usually the case, but the operating point can shift it beyond 30° ; therefore, it is necessary to verify graphically or analytically where the maximum current occurs and then substitute that instant to get the peak armature current from the following equation:

$$i_a(t) = \left(\frac{V_m}{|Z_a|}\right)\left\{\sin(\omega_s t + \pi/3 + \alpha - \beta) - \sin(\pi/3 + \alpha - \beta)e^{-t/T_s}\right\} - \left(\frac{E}{R_a}\right)(1 - e^{-t/T_s}) + i_{a1}e^{-t/T_s} = 2411.5 \text{ A}$$

The armature current ripple magnitude, $\Delta i_a = 2411.5 - 2308.1 = 103.4 \text{ A}$

The ripple torque magnitude, $\Delta T_e = K_b \Delta i_a = 2.646 * 103.4 = 273.86 \text{ N}\cdot\text{m}$

Average air gap torque, $T_{e(av)} = I_{av} K_b \cong [(2411.5 + 2308.1) * 0.5] * 2.646 = 6244 \text{ N}\cdot\text{m}$

Note that the ripple current magnitude is less than 5% and therefore is approximated as a straight line between its minimum and maximum values in each part of its cycle.

Torque ripple as a percent of the operating average torque is

$$\Delta T_{en} = \frac{\Delta T_e}{T_{e(av)}} * 100 = \frac{273.86}{6244} * 100 = 4.4\%$$

3.5 Two-Quadrant Three-Phase Converter-Controlled DC Motor Drive

The control schematic of a two-quadrant converter-controlled separately-excited dc motor drive is shown in Figure 3.26. The motor drive shown is a speed-controlled system. The thyristor bridge converter gets its ac supply through a three-phase transformer and fast-acting ac contactors. The dc output is fed to the armature of the dc motor. The field is separately excited, and the field supply can be kept constant or regulated, depending on the need for the field-weakening mode of operation. The dc motor has a tachogenerator whose output is utilized for closing the speed loop. The motor is driving a load considered to be frictional for this treatment. The output of the tachogenerator is filtered to remove the ripples to provide the signal, ω_{mr} . The speed command ω_r^* is compared to the speed signal to produce a speed error signal. This signal is processed through a proportional-plus-integral (PI) controller to determine the torque command. The torque command is limited, to keep it within the safe

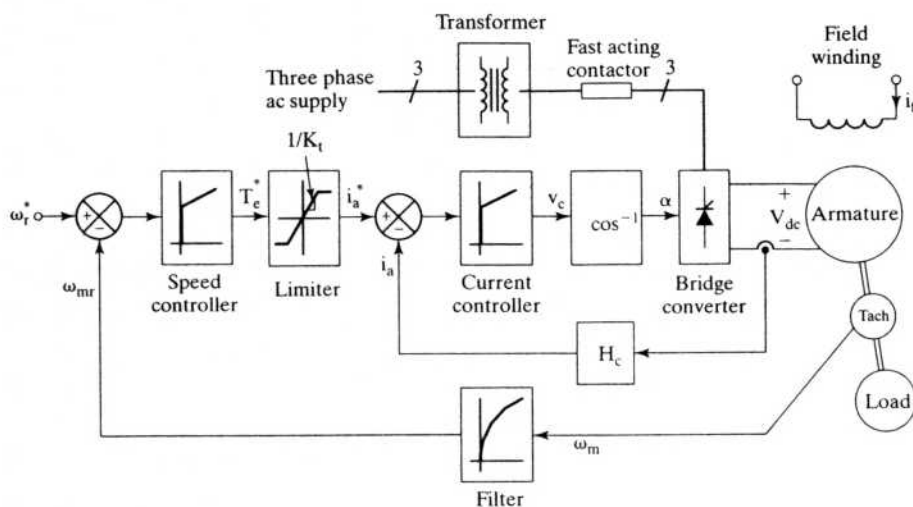


Figure 3.26 Speed-controlled two-quadrant dc motor drive

current limits, and the current command is obtained by proper scaling. The armature current command i_a^* is compared to the actual armature current i_a to have a zero current error. In case there is an error, a PI current controller processes it to alter the control signal v_c . The control signal accordingly modifies the triggering angle α to be sent to the converter for implementation. The implementation of v_c to α in the converter is discussed under control circuit in section 3.3.3.

The inner current loop assures a fast current response and hence also limits the current to a safe preset level. This inner current loop makes the converter a linear current amplifier. The outer speed loop ensures that the actual speed is always equal to the commanded speed and that any transient is overcome within the shortest feasible time without exceeding the motor and converter capability.

The operation of the closed-loop speed-controlled drive is explained from one or two particular instances of speed command. A speed from zero to rated value is commanded, and the motor is assumed to be at standstill. This will generate a large speed error and a torque command and in turn an armature current command. The armature current error will generate the triggering angle to supply a preset maximum dc voltage across the motor terminals. The inner current loop will maintain the current at the level permitted by its commanded value, producing a corresponding torque. As the motor starts running, the torque and current are maintained at their maximum level, thus accelerating the motor rapidly. When the rotor attains the commanded value, the torque command will settle down to a value equal to the sum of the load torque and other motor losses to keep the motor in steady state.

The design of the gain and time constants of the speed and current controllers is of paramount importance in meeting the dynamic specifications of the motor drive. Their systematic designs are considered in the next section.

3.6 TRANSFER FUNCTIONS OF THE SUBSYSTEMS

3.6.1 DC Motor and Load

The dc machine contains an inner loop due to the induced emf. It is not physically seen; it is magnetically coupled. The inner current loop will cross this back-emf loop, creating a complexity in the development of the model. It is shown in Figure 3.27. The interactions of these loops can be decoupled by suitably redrawing the block diagram. The development of such a block diagram for the dc machine is shown in Figure 3.28, step by step. The load is assumed to be proportional to speed and is given as

$$T_l = B_l \omega_m \quad (3.61)$$

To decouple the inner current loop from the machine-inherent induced-emf loop, it is necessary to split the transfer function between the speed and voltage into two cascade transfer functions, first between speed and armature current and then between armature current and input voltage, represented as

$$\frac{\omega_m(s)}{V_a(s)} = \frac{\omega_m(s)}{I_a(s)} \cdot \frac{I_a(s)}{V_a(s)} \quad (3.62)$$

where

$$\frac{\omega_m(s)}{I_a(s)} = \frac{K_b}{B_l(1 + sT_m)} \quad (3.63)$$

$$\frac{I_a(s)}{V_a(s)} = K_l \frac{(1 + sT_m)}{(1 + sT_1)(1 + sT_2)} \quad (3.64)$$

$$T_m = \frac{J}{B_l} \quad (3.65)$$

$$B_l = B_1 + B_l \quad (3.66)$$

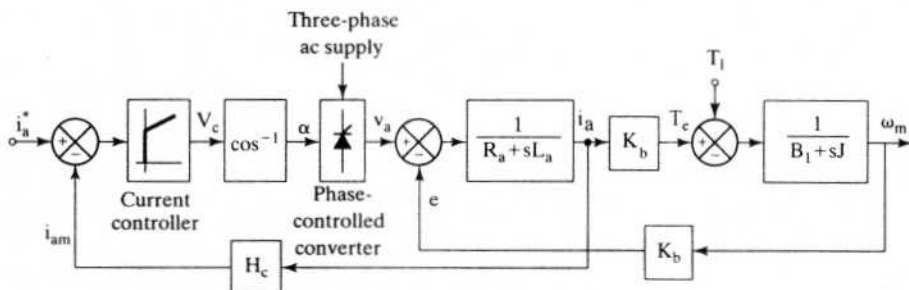
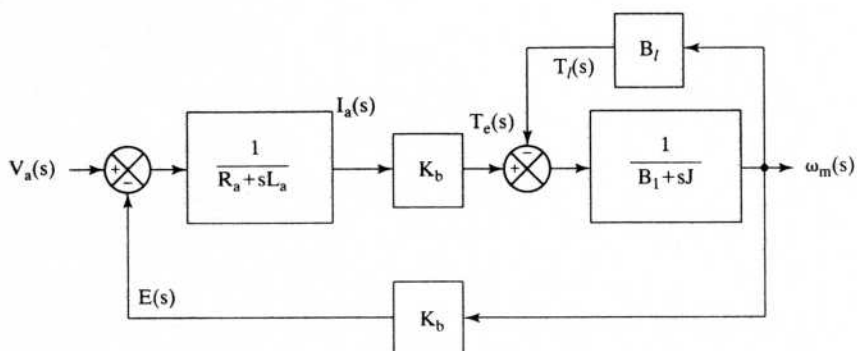
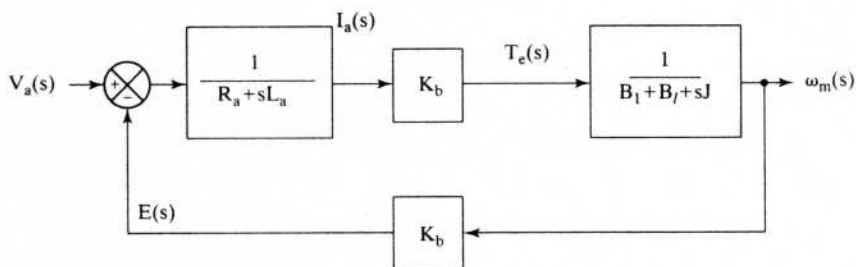


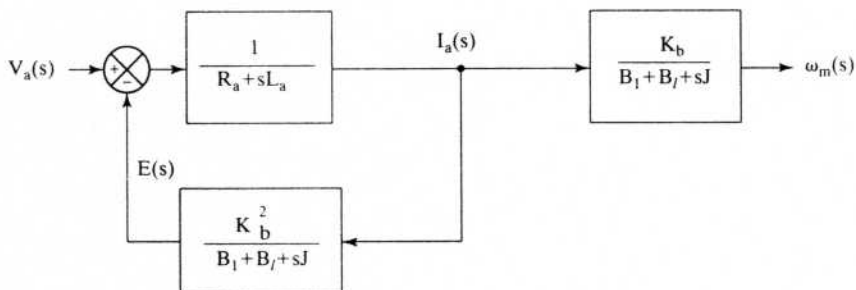
Figure 3.27 DC motor and current-control loop



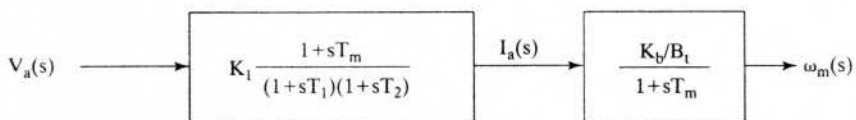
Step 1



Step 2



Step 3



Step 4

$$-\frac{1}{T_1}, -\frac{1}{T_2} = -\frac{1}{2} \left[\frac{B_t}{J} + \frac{R_a}{L_a} \right] \pm \sqrt{\frac{1}{4} \left(\frac{B_t}{J} + \frac{R_a}{L_a} \right)^2 - \left(\frac{K_b^2 + R_a B_t}{J L_a} \right)} \quad (3.67)$$

$$K_1 = \frac{B_t}{K_b + R_a B_t} \quad (3.68)$$

3.6.2 Converter

The converter after linearization is represented as

$$G_r(s) = \frac{V_a(s)}{v_c(s)} = \frac{K_r}{1 + sT_r} \quad (3.69)$$

The delay time T_r and gain are evaluated and given in section 3.3.4.

3.6.3 Current and Speed Controllers

The current and speed controllers are of proportional-integral type. They are represented as

$$G_c(s) = \frac{K_c(1 + sT_c)}{sT_c} \quad (3.70)$$

$$G_s(s) = \frac{K_s(1 + sT_s)}{sT_s} \quad (3.71)$$

where the subscripts c and s correspond to the current and speed controllers, respectively. The K and T correspond to the gain and time constants of the controllers.

3.6.4 Current Feedback

The gain of the current feedback is H_c . No filtering is required in most cases. In the case of a filtering requirement, a low-pass filter can be included in the analysis. Even then, the time constant of the filter must not be greater than a millisecond.

3.6.5 Speed Feedback

Most high performance systems use a dc tachogenerator, and the filter required is low-pass, with a time constant under 10 ms. The transfer function of the speed feedback filter is

$$G_w(s) = \frac{K_w}{1 + sT_w} \quad (3.72)$$

where K_w is the gain and T_w is the time constant.

3.7
DESIGN OF CONTROLLERS

The overall closed-loop system is shown in Figure 3.29. It is seen that the current loop does not contain the inner induced-emf loop. The design of control loops starts from the innermost (fastest) loop and proceeds to the slowest loop, which in this case is the outer speed loop. The reason to proceed from the inner to the outer loop in the design process is that the gain and time constants of only one controller at a time are solved, instead of solving for the gain and time constants of all the controllers simultaneously. Not only is that logical; it also has a practical implication. Note that every motor drive need not be speed-controlled but may be torque-controlled, such as for a traction application. In that case, the current loop is essential and exists regardless of whether the speed loop is going to be closed. Additionally, the performance of the outer loop is dependent on the inner loop; therefore, the tuning of the inner loop has to precede the design and tuning of the outer loop. That way, the dynamics of the inner loop can be simplified and the impact of the outer loop on its performance could be minimized. The design of the current and speed controllers is considered in this section.

3.7.1
Current Controller

The current-control loop is shown in Figure 3.30. The loop gain function is

$$GH_i(s) = \left\{ \frac{K_i K_c K_r H_c}{T_c} \right\} \cdot \frac{(1 + sT_c)(1 + sT_m)}{s(1 + sT_1)(1 + sT_2)(1 + sT_r)} \tag{3.73}$$

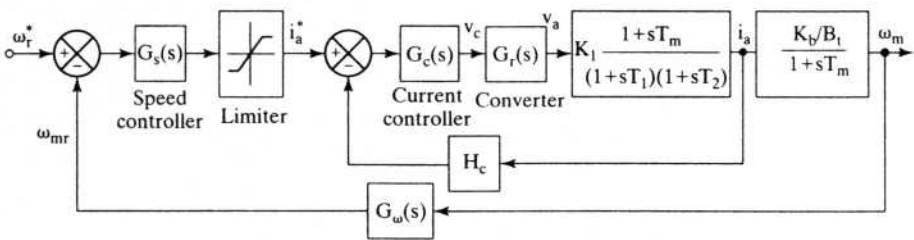


Figure 3.29
 Block diagram of the motor drive

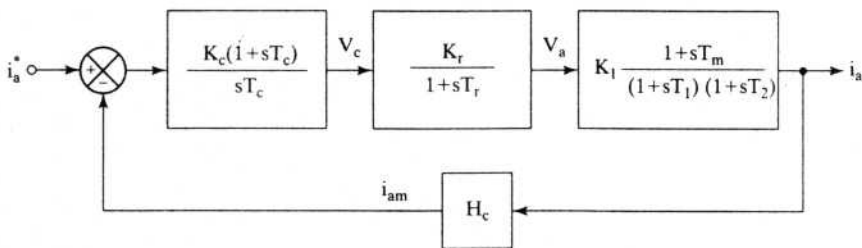


Figure 3.30
 Current-control loop

This is a fourth-order system, and simplification is necessary to synthesize a controller without resorting to a computer. Noting that T_m is on the order of a second and in the vicinity of the gain crossover frequency, we see that the following approximation is valid:

$$(1 + sT_m) \cong sT_m \quad (3.74)$$

which reduces the loop gain function to

$$GH_i(s) \cong \frac{K(1 + sT_c)}{(1 + sT_1)(1 + sT_2)(1 + sT_r)} \quad (3.75)$$

where

$$K = \frac{K_1 K_c K_r H_c T_m}{T_c} \quad (3.76)$$

The time constants in the denominator are seen to have the relationship

$$T_r < T_2 < T_1 \quad (3.77)$$

The equation (3.75) can be reduced to second order, to facilitate a simple controller synthesis, by judiciously selecting

$$T_c = T_2 \quad (3.78)$$

Then the loop function is

$$GH_i(s) \cong \frac{K}{(1 + sT_1)(1 + sT_r)} \quad (3.79)$$

The characteristic equation or denominator of the transfer function between the armature current and its command is

$$(1 + sT_1)(1 + sT_r) + K \quad (3.80)$$

This equation is expressed in standard form as

$$T_1 T_r \left\{ s^2 + s \left(\frac{T_1 + T_r}{T_1 T_r} \right) + \frac{K + 1}{T_1 T_r} \right\} \quad (3.81)$$

from which the natural frequency and damping ratio are obtained as

$$\omega_n^2 = \frac{K + 1}{T_1 T_r} \quad (3.82)$$

$$\zeta = \frac{\left(\frac{T_1 + T_r}{T_1 T_r} \right)}{2 \sqrt{\frac{K + 1}{T_1 T_r}}} \quad (3.83)$$

where ω_n and ζ are the natural frequency and damping ratio, respectively. For good dynamic performance, it is an accepted practice to have a damping ratio of 0.707. Hence, equating the damping ratio to 0.707 in equation (3.83), we get

$$K + 1 = \frac{\left(\frac{T_1 + T_r}{T_1 T_r}\right)^2}{\left(\frac{2}{T_1 T_r}\right)} \quad (3.84)$$

Realizing that

$$K \gg 1 \quad (3.85)$$

$$T_1 \gg T_r \quad (3.86)$$

tells us that K is approximated as

$$K \cong \frac{T_1^2}{2T_1 T_r} \cong \frac{T_1}{2T_r} \quad (3.87)$$

By equating (3.76) to (3.87), the current-controller gain is evaluated as

$$K_c = \frac{1}{2} \cdot \frac{T_1 T_c}{T_r} \cdot \left(\frac{1}{K_1 K_r H_c T_m}\right) \quad (3.88)$$

3.7.2 First-Order Approximation of Inner Current Loop

To design the speed loop, the second-order model of the current loop is replaced with an approximate first-order model. This helps to reduce the order of the overall speed-loop gain function. The current loop is approximated by adding the time delay in the converter block to T_1 of the motor; because of the cancellation of one motor pole by a zero of the current controller, the resulting current loop can be shown in Figure 3.31. The transfer function of the current and its commanded value is

$$\frac{I_a(s)}{I_a^*(s)} = \frac{\frac{K_c K_r T_1 T_m}{T_c} \cdot \frac{1}{(1 + sT_3)}}{1 + \frac{K_1 K_c K_r H_c T_m}{T_c} \cdot \frac{1}{(1 + sT_3)}} \quad (3.89)$$

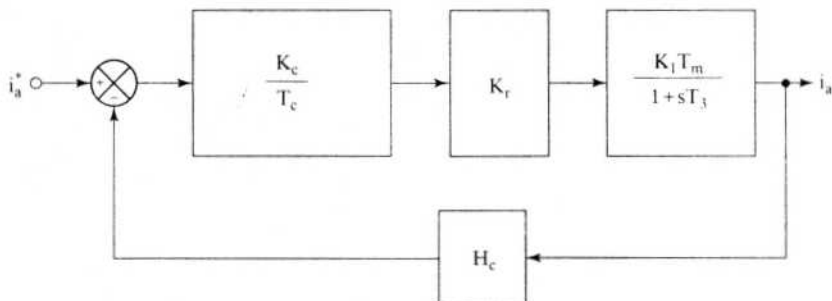


Figure 3.31 Simplified current-control loop

where $T_3 = T_l + T_r$. The transfer function can be arranged simply as

$$\frac{I_a(s)}{I_a^*(s)} = \frac{K_i}{(1 + sT_i)} \quad (3.90)$$

where

$$T_i = \frac{T_3}{1 + K_{fi}} \quad (3.91)$$

$$K_i = \frac{K_{fi}}{H_c} \cdot \frac{1}{(1 + K_{fi})} \quad (3.92)$$

$$K_{fi} = \frac{K_c K_r K_l T_m H_c}{T_c} \quad (3.93)$$

The resulting model of the current loop is a first-order system, suitable for use in the design of a speed loop. The gain and delay of the current loop can also be found experimentally in a motor-drive system. That would be more accurate for the speed-controller design.

3.7.3 Speed Controller

The speed loop with the first-order approximation of the current-control loop is shown in Figure 3.32. The loop gain function is

$$GH_s(s) = \left\{ \frac{K_s K_i K_b H_w}{B_l T_s} \right\} \cdot \frac{(1 + sT_s)}{s(1 + sT_i)(1 + sT_m)(1 + sT_w)} \quad (3.94)$$

This is a fourth-order system. To reduce the order of the system for analytical design of the speed controller, approximation serves. In the vicinity of the gain crossover frequency, the following is valid:

$$(1 + sT_m) \cong sT_m \quad (3.95)$$

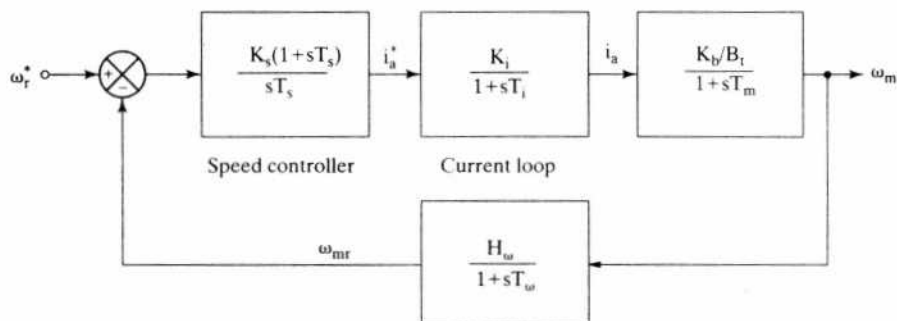


Figure 3.32 Representation of the outer speed loop in the dc motor drive

The next approximation is to build the equivalent time delay of the speed feedback filter and current loop. Their sum is very much less than the integrator time constant, T_s , and hence the equivalent time delay, T_4 , can be considered the sum of the two delays, T_i and T_w . This step is very similar to the equivalent time delay introduced in the simplification of the current-loop transfer function. Hence, the approximate gain function of the speed loop is

$$GH_s(s) \cong K_2 \cdot \frac{K_s}{T_s} \cdot \frac{(1 + sT_s)}{s^2(1 + sT_4)} \quad (3.96)$$

where

$$T_4 = T_i + T_w \quad (3.97)$$

$$K_2 = \frac{K_f K_b H_w}{B_t T_m} \quad (3.98)$$

The closed-loop transfer function of the speed to its command is

$$\frac{\omega_m(s)}{\omega_r^*(s)} = \frac{1}{H_w} \left[\frac{\frac{K_2 K_s}{T_s} (1 + sT_s)}{s^3 T_4 + s^2 + s K_2 K_s + \frac{K_2 K_s}{T_s}} \right] = \frac{1}{H_w} \frac{(a_0 + a_1 s)}{(a_0 + a_1 s + a_2 s^2 + a_3 s^3)} \quad (3.99)$$

where

$$a_0 = K_2 K_s / T_s \quad (3.100)$$

$$a_1 = K_2 K_s \quad (3.101)$$

$$a_2 = 1 \quad (3.102)$$

$$a_3 = T_4 \quad (3.103)$$

This transfer function is optimized to have a wider bandwidth and a magnitude of one over a wide frequency range by looking at its frequency response. Its magnitude is given by

$$\left| \frac{\omega_m(j\omega)}{\omega_r^*(j\omega)} \right| = \frac{1}{H_w} \sqrt{\frac{a_0^2 + \omega^2 a_1^2}{\{a_0^2 + \omega^2(a_1^2 - 2a_0 a_2) + \omega^4(a_2^2 - 2a_1 a_3) + \omega^6 a_3^2\}}} \quad (3.104)$$

This is optimized by making the coefficients of ω^2 and ω^4 equal zero, to yield the following conditions:

$$a_1^2 = 2a_0 a_2 \quad (3.105)$$

$$a_2^2 = 2a_1 a_3 \quad (3.106)$$

Substituting these conditions in terms of the motor and controller parameters given in (3.100) into (3.103) yields

$$T_s^2 = \frac{2T_s}{K_s K_2} \quad (3.107)$$

resulting in

$$T_s K_s = \frac{2}{K_2} \quad (3.108)$$

Similarly,

$$\frac{T_s^2}{K_s^2 K_2^2} = \frac{2T_s^2 T_4}{K_s K_2} \quad (3.109)$$

which, after simplification, gives the speed-controller gain as

$$K_s = \frac{1}{2K_2 T_4} \quad (3.110)$$

Substituting equation (3.110) into equation (3.108) gives the time constant of the speed controller as

$$T_s = 4T_4 \quad (3.111)$$

Substituting for K_s and T_s into (3.99) gives the closed-loop transfer function of the speed to its command as

$$\frac{\omega_m(s)}{\omega_r^*(s)} = \frac{1}{H_w} \left[\frac{1 + 4T_4 s}{1 + 4T_4 s + 8T_4^2 s^2 + 8T_4^3 s^3} \right] \quad (3.112)$$

It is easy to prove that for the open-loop gain function the corner points are $1/4T_4$ and $1/T_4$, with the gain crossover frequency being $1/2T_4$. In the vicinity of the gain crossover frequency, the slope of the magnitude response is -20 dB/decade, which is the most desirable characteristic for good dynamic behavior. Because of its symmetry at the gain crossover frequency, this transfer function is known as a symmetric optimum function. Further, this transfer function has the following features:

- (i) Approximate time constant of the system is $4T_4$.
- (ii) The step response is given by

$$\omega_r(t) = \frac{1}{H_w} (1 + e^{-t/2T_4} - 2e^{-t/4T_4} \cos(\sqrt{3}t/4T_4)) \quad (3.113)$$

with a rise time of $3.1T_4$, a maximum overshoot of 43.4%, and a settling time of $16.5T_4$.

- (iii) Since the overshoot is high, it can be reduced by compensating for its cause, i.e., the zero of a pole in the speed command path, as shown in Figure 3.33. The resulting transfer function of the speed to its command is

$$\frac{\omega_m(s)}{\omega_r^*(s)} = \frac{1}{H_w} \left[\frac{1}{1 + 4T_4 s + 8T_4^2 s^2 + 8T_4^3 s^3} \right] \quad (3.114)$$

whose step response is

$$\omega_r(t) = \frac{1}{H_w} \left(1 - e^{-t/4T_4} - \frac{2}{\sqrt{3}} e^{-t/4T_4} \sin(\sqrt{3}t/4T_4) \right) \quad (3.115)$$

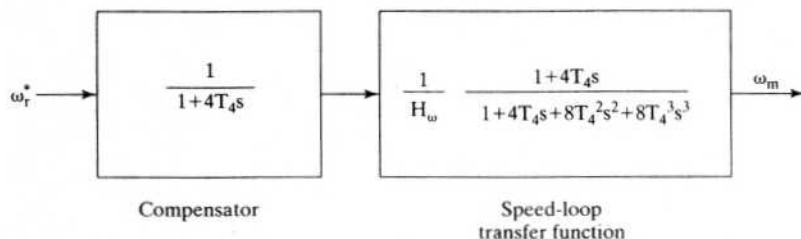


Figure 3.33 Smoothing of the overshoot via a compensator

with a rise time of $7.6T_4$, a maximum overshoot of 8.1%, and a settling time of $13.3T_4$. Even though the rise time has increased, the overshoot has been reduced to approximately 20% of its previous value, and the settling time has come down by 19%.

(iv) The poles of the closed-loop transfer function are

$$s = -\frac{1}{2T_4}, -\frac{1}{4T_4} \pm j\frac{\sqrt{3}}{4T_4} \quad (3.116)$$

The real parts of the poles are negative, and there are no repeated poles at the origin, so the system is asymptotically stable. Hence, in the symmetric optimum design, the system stability is guaranteed, and there is no need to check for it in the design process. Whether this is true for the original system without approximation will be explored in the following example.

(v) Symmetric optimum eliminates the effects due to the disturbance very rapidly compared to other optimum techniques employed in practical systems, such as linear or modulus optimum. This approach indicates one of the possible methods to synthesize the speed controller. That the judicious choice of approximation is based on the physical constants of the motor, on the converter and transducer gains, and on time delays is to be emphasized here.

That the speed-loop transfer function is expressed in terms of T_4 is significant in that it clearly links the dynamic performance to the speed-feedback and current-loop time constants. That a faster current loop with a smaller speed-filter time constant accelerates the speed response is evident from this. Expressing T_4 in terms of the motor, the converter and transducer gains, and the time delays by using expressions (3.91) and (3.97) yields

$$T_4 = T_i + T_w = \frac{T_3}{1 + K_{fi}} + T_w = \frac{T_i + T_r}{1 + K_{fi}} + T_w \quad (3.117)$$

Since $K_{fi} \gg 1$, T_4 is found approximately after substituting for K_{fi} from equation (3.93) in terms of gains and time delays as

$$T_4 \approx \frac{(T_i + T_r)T_2}{T_m} \cdot \frac{1}{K_l K_c K_r H_c} + T_w \quad (3.118)$$

This clearly shows the influence of the subsystem parameters on the system dynamics. A clear understanding of this would help the proper selection of the subsystems

to obtain the required dynamic performance of the speed-controlled motor-drive system. Further, this derivation demonstrates that the system behavior to a large degree depends on the subsystem parameters rather than only on the current and speed-controller parameters or on the sophistication of their design.

Example 3.6

Design a speed-controlled dc motor drive maintaining the field flux constant. The motor parameters and ratings are as follows:

220 V, 8.3 A, 1470 rpm, $R_a = 4 \Omega$, $J = 0.0607 \text{ kg} \cdot \text{m}^2$, $L_a = 0.072 \text{ H}$, $B_f = 0.0869 \text{ N} \cdot \text{m} / \text{rad} \cdot \text{sec}$, $K_b = 1.26 \text{ V} / \text{rad} \cdot \text{sec}$.

The converter is supplied from 230V, 3-phase ac at 60 Hz. The converter is linear, and its maximum control input voltage is $\pm 10 \text{ V}$. The tachogenerator has the transfer function

$G_w(s) = \frac{0.065}{(1 + 0.002s)}$. The speed reference voltage has a maximum of 10V. The maximum current permitted in the motor is 20 A.

Solution (i) Converter transfer function:

$$K_r = \frac{1.35 \text{ V}}{V_{cm}} = \frac{1.35 \times 230}{10} = 31.05 \text{ V/V}$$

$$V_{dc}(\text{max}) = 310.5 \text{ V}$$

The rated dc voltage required is 220 V, which corresponds to a control voltage of 7.09 V. The transfer function of the converter is

$$G_r(s) = \frac{31.05}{(1 + 0.00138s)} \text{ V/V}$$

(ii) Current transducer gain: The maximum safe control voltage is 7.09 V, and this has to correspond to the maximum current error:

$$i_{\max} = 20 \text{ A}$$

$$H_c = \frac{7.09}{I_{\max}} = \frac{7.09}{20} = 0.355 \text{ V/A}$$

(iii) Motor transfer function:

$$K_1 = \frac{B_f}{K_b^2 + R_a B_f} = \frac{0.0869}{1.26^2 + 4 \times 0.0869} = 0.0449$$

$$-\frac{1}{T_1} - \frac{1}{T_2} = -\frac{1}{2} \left[\frac{B_f}{J} + \frac{R_a}{L_a} \right] \pm \sqrt{\frac{1}{4} \left(\frac{B_f}{J} + \frac{R_a}{L_a} \right)^2 - \left(\frac{K_b^2 + R_a B_f}{J L_a} \right)}$$

$$T_1 = 0.1077 \text{ sec}$$

$$T_2 = 0.0208 \text{ sec}$$

$$T_m = \frac{J}{B_f} = 0.7 \text{ sec}$$

The subsystem transfer functions are

$$\frac{I_a(s)}{V_a(s)} = K_1 \frac{(1 + sT_m)}{(1 + sT_1)(1 + sT_2)} = \frac{0.0449(1 + 0.7s)}{(1 + 0.0208s)(1 + 0.1077s)}$$

$$\frac{\omega_m(s)}{I_a(s)} = \frac{K_b/B_t}{(1 + sT_m)} = \frac{14.5}{(1 + 0.7s)}$$

(iv) Design of current controller:

$$T_c = T_2 = 0.0208 \text{ sec}$$

$$K = \frac{T_1}{2T_r} = \frac{0.1077}{2 \times 0.001388} = 38.8$$

$$K_c = \frac{KT_c}{K_1 H_c K_r T_m} = \frac{38.8 \times 0.0208}{0.0449 \times 0.355 \times 31.05 \times 0.7} = 2.33$$

(v) Current-loop approximation:

$$\frac{I_a(s)}{I_a^*(s)} = \frac{K_i}{(1 + sT_i)}$$

where

$$K_i = \frac{K_{fi}}{H_c} \cdot \frac{1}{(1 + K_{fi})}$$

$$K_{fi} = \frac{K_c K_r K_1 T_m H_c}{T_c} = 38.8$$

$$\therefore K_i = \frac{27.15}{28.09} \cdot \frac{1}{0.355} = 2.75$$

$$T_i = \frac{T_3}{1 + K_{fi}} = \frac{0.109}{1 + 38.8} = 0.0027 \text{ sec}$$

The validity of the approximations is evaluated by plotting the frequency response of the closed-loop current to its command, with and without the approximations. This is shown in Figure 3.34. From this figure, it is evident that the approximations are quite valid in the frequency range of interest.

(vi) Speed-controller design:

$$T_4 = T_i + T_w = 0.0027 + 0.002 = 0.0047 \text{ sec}$$

$$K_2 = \frac{K_i K_b H_w}{B_r T_m} = \frac{2.75 \times 1.26 \times 0.065}{0.0869 \times 0.7} = 3.70$$

$$K_s = \frac{1}{2K_2 T_4} = \frac{1}{2 \times 3.70 \times 0.0047} = 28.73$$

$$T_s = 4T_4 = 4 \times 0.0047 = 0.0188 = \text{sec}$$

The frequency responses of the speed to its command are shown in Figure 3.35 for cases with and without approximations. That the model reduction with the approximations has given a transfer function very close to the original is obvious from this figure. Further, the

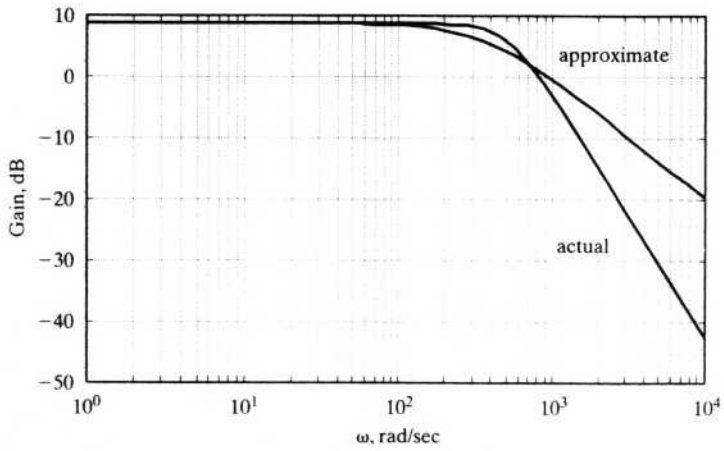


Figure 3.34 Frequency response of the current-transfer functions with and without approximation

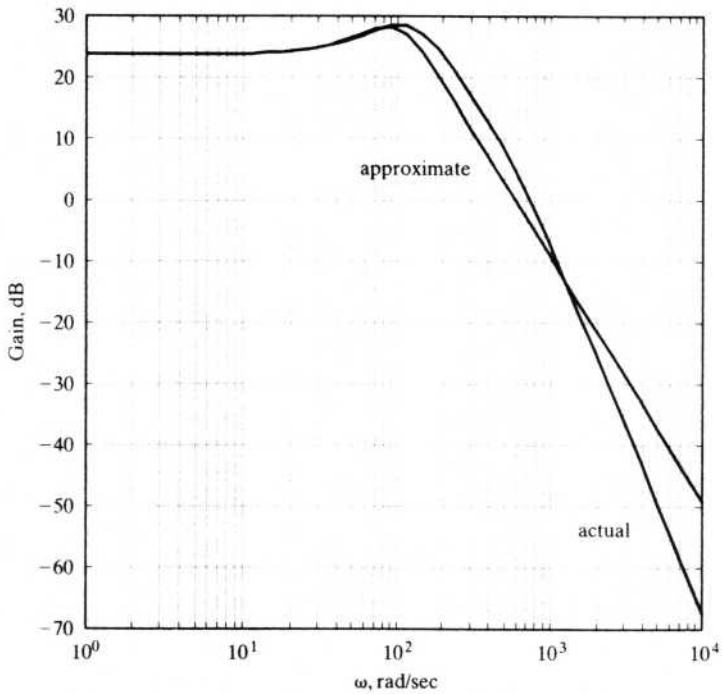


Figure 3.35 Frequency response of the speed-transfer functions with and without approximation

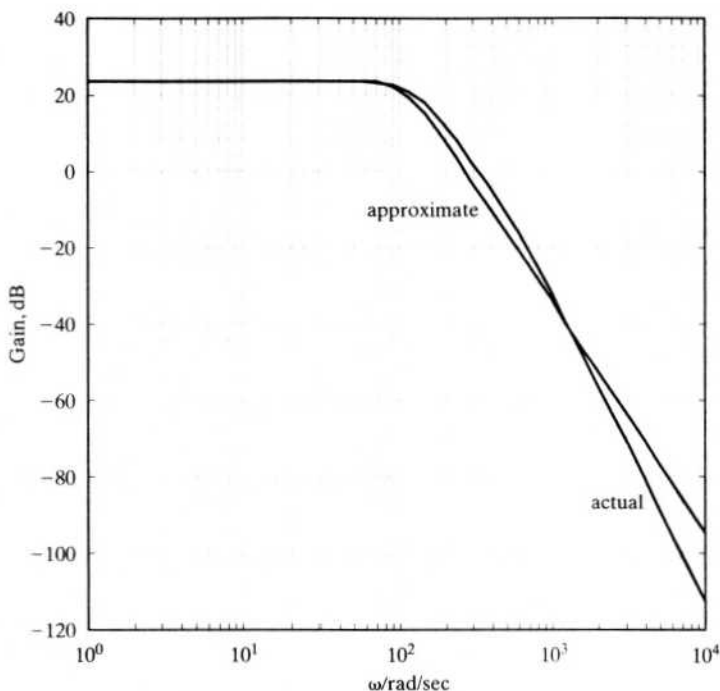


Figure 3.36 Frequency response of the speed-transfer function with smoothing for the cases with and without approximation

smoothing of the overshoot by the cancellation of the zero with a pole at $-1/4T_4$ is shown in Figure 3.36. This figure contains the approximated transfer function of third order for the speed to its command-transfer function and the one without any approximations. Again, the closeness of these two solutions justifies the approximations.

The time responses are important to verify the design of the controllers, and they are shown in Figure 3.37 for the case without smoothing and with smoothing. The case without any approximation is included here for the comparison of all responses.

Example 3.7

Assume motor poles are complex. Develop a design procedure for the current controller.

Solution If motor poles are complex, then the procedure outlined above is not applicable for the design of the current and speed controllers. One alternative is as follows: the current controller is designed by using the symmetric optimum criterion that was applied in the earlier speed controller design. The steps are given below.

Assuming $(1 + sT_m) \cong sT_m$ leads to the following current-loop transfer function:

$$\frac{i_a(s)}{i_a^*(s)} = \frac{K_c \frac{K_z}{T_c} (1 + sT_c)}{b_0 + b_1s + b_2s^2 + b_3s^3}$$

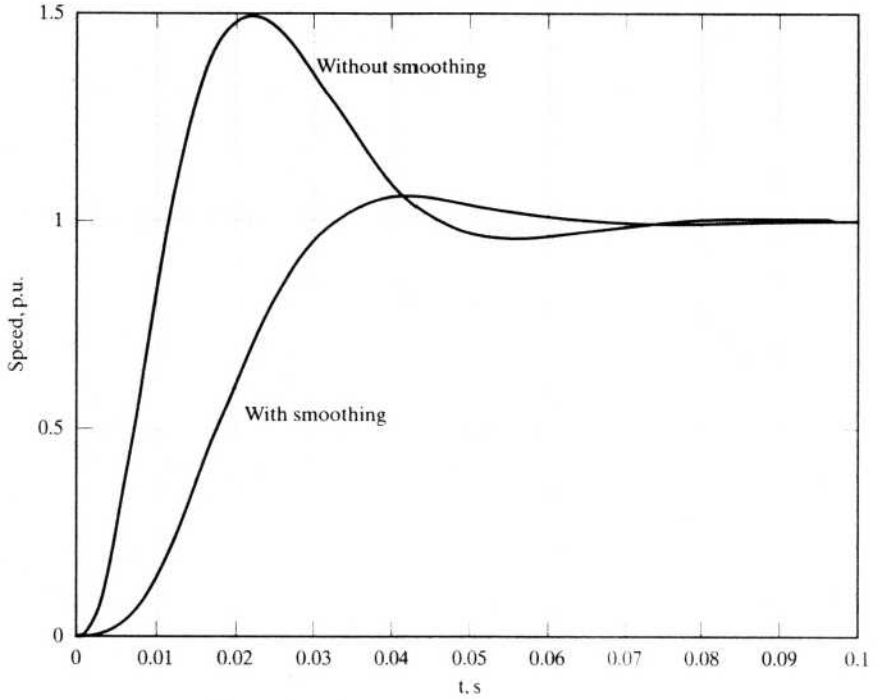


Figure 3.37 Time response of the speed controller

where

$$K_2 = K_1 K_r T_m$$

$$b_0 = 1 + K_2 \frac{K_c}{T_c} H_c$$

$$b_1 = T_1 + T_2 + T_r + K_2 K_c H_c$$

$$b_2 = (T_1 + T_2)T_r + T_1 T_2$$

$$b_3 = T_1 T_2 T_r$$

Applying symmetric-optimum conditions,

$$b_1^2 = 2b_0 b_2$$

$$(T_1 + T_2 + T_r + K_2 K_c H_c)^2 = 2 \left(1 + K_2 \frac{K_c}{T_c} H_c \right) ((T_1 + T_2)T_r + T_1 T_2)$$

$$b_2^2 = 2b_1 b_3$$

$$((T_1 + T_2)T_r + T_1 T_2)^2 = 2(T_1 + T_2 + T_r + K_2 K_c H_c)(T_1 T_2 T_r)$$

but $T_r \ll T_1, T_2$,

$$\begin{aligned}
&\therefore (T_1 + T_2)T_r < T_1T_2 \\
&\frac{(T_1T_2)^2}{2(T_1T_2T_r)} = T_1 + T_2 + T_r + K_2K_cH_c \\
&\frac{T_1T_2}{2T_r} \cong K_2K_cH_c \left[\because T_1 + T_2 + T_r < \frac{T_1T_2}{2T_r} \right] \\
&\therefore K_c = \frac{T_1T_2}{2T_r} \frac{1}{K_2H_c}
\end{aligned}$$

Also,

$$\begin{aligned}
\left(T_1 + T_2 + T_r + \frac{T_1T_2}{2T_r} \right)^2 &= 2 \left(1 + \frac{T_1T_2}{2T_rT_c} \right) ((T_1 + T_2)T_r + T_1T_2) \\
\frac{(T_1T_2)^2}{4T_r^2} &\cong 2T_1T_2 \left(1 + \frac{T_1T_2}{2T_rT_c} \right) \\
\frac{1}{4T_r^2} &\cong \frac{1}{T_rT_c} \\
T_c &\cong \frac{4T_r^2}{T_r} \cong 4T_r
\end{aligned}$$

The next step is to obtain the first-order approximation of the current-loop transfer function for the synthesis of the speed controller. Since the time constant T_c is known, the first-order approximation of the current loop is written as

$$\frac{i_a(s)}{i_a'(s)} \cong \frac{K_i}{1 + sT_i}$$

where the steady-state gain is obtained from the exact transfer function, by setting $s = 0$, as

$$K_i = \frac{\frac{K_2K_c}{T_c}}{1 + \frac{K_2K_cH_c}{T_c}}$$

and $T_i = T_c$.

From this point, the speed-controller design follows the symmetric-optimum procedure outlined earlier for the case with the real motor poles.

3.8 TWO-QUADRANT DC MOTOR DRIVE WITH FIELD WEAKENING

Flux weakening is exercised in the machine by varying the voltage applied to the field winding. The applied voltage is varied via a single-phase or three-phase half-wave controller rectifier. Because of the large inductance of the field winding, the field current is smooth and hardly has ripple. The converter time lag is negligible compared to the field time constant, and the speed of response is determined primarily by the field time constant. Therefore, there is very little justification for using full-wave converters in motor drives of rating less than 100 kW. A schematic of the two-quadrant dc motor drive with flux weakening is shown in Figure 3.38.

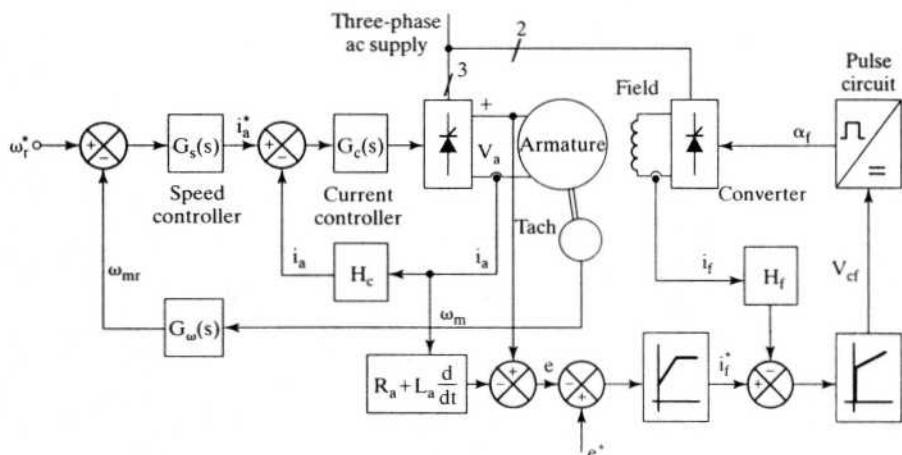


Figure 3.38 A two-quadrant dc motor drive with field weakening

The command field current is determined by the induced-emf error function. The induced-emf error is the difference between the reference induced emf and the estimated machine-induced emf. The machine-induced emf is found by subtracting the resistive and inductive drop from the applied voltage. This induced-emf estimator is machine-parameter sensitive and needs to be adaptive. The induced-emf error is amplified by a PI controller and limited to yield the field-current reference. The field-current reference is enforced by a current-control loop feedback very similar to the armature current-control loop. The outer induced-emf feedback loop enforces a constant induced emf for speeds higher than the base speed. This amounts to

$$e_n = \phi_{fn} \omega_{mn} \quad (3.119)$$

If the induced emf e_n is kept at rated value, say 1 p.u., then the field flux is inversely proportional to the rotor speed. This condition also enables constant-air gap-power operation.

3.9 FOUR-QUADRANT DC MOTOR DRIVE

A four-quadrant dc motor drive has a set of dual three-phase converters for the power stage. Its control is very similar to that of the two-quadrant dc motor drive. Converters have to be energized depending on the quadrant of operation. Converters 1 and 2 are for forward and reverse directions of rotation of the motor, respectively. The changeover from one converter to another is safely handled by monitoring speed, current-command, and zero-crossing current signals. These signals form the inputs to the selector block, which assigns the pulse-control signals to the appropriate converter. The converters share the same current and speed loops. A control schematic of the four-quadrant dc motor drive is shown in Figure 3.39. The selector block will switch the converters over only when the current in the outgoing

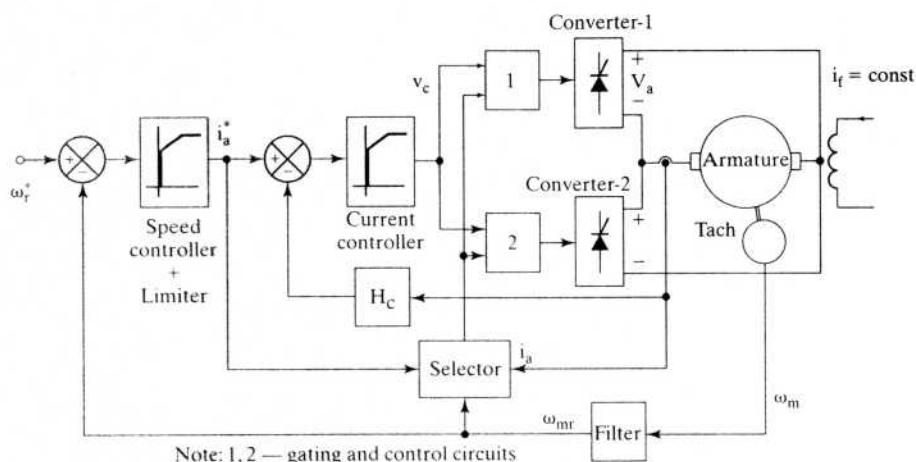


Figure 3.39 A four-quadrant dc motor drive

converter has come to zero. Apart from that, some other conditions have to be satisfied to transfer the control of converters. Assuming that a positive speed command is required for forward running and a negative speed command is required for reverse running, the crossover control for converters is discussed. The machine is running at rated speed and the speed command is changed to reverse rated speed. The current command becomes negative, indicating that the machine has to become regenerative in order to decelerate the motor in the forward direction. Forward regeneration is possible only in quadrant IV, and the converter to provide this quadrant of operation is converter 2. Before the current is reversed, it has to go through zero. That is achieved by increasing the triggering angle of converter 1. When the current is zero, a certain dead time is given to enable the thyristors in converter 1 to recover reverse blocking capability. After this interval, converter 2 is enabled. At this time, the armature current has been forced to zero, but the speed is still positive. The triggering angle of converter 2 is set such that its output voltage equals and opposes the induced emf. Then, slowly decreasing the triggering angle increases the armature current in the opposite direction. By a varying of the triggering angle, the armature current is fully reversed and then is maintained at the reference level. When the motor reaches zero speed, the operation of converter 2 is continued by bringing it into rectification mode, i.e., with α less than 90° . That will accelerate the motor in the reverse direction until it matches the speed reference.

The function of the selector block is to determine which converter has to be operating. In the previous discussion, as soon as the current command goes negative, the selector block will transfer the control from converter 1 to converter 2 with proper initial triggering angle. If the triggering angle of converter 1 is α_1 , then the initial triggering angle, α_2 , of converter 2 is $(180^\circ - \alpha_1)$, to match the output voltage of converter 1. If circulating current is not allowed between the converters, zero crossing of the armature current is required to transfer the control from one converter to another. The actual rotor speed is required to determine the quadrant of

operation. Based on the rotor speed, the armature current, and its command, the selector block identifies the converter for control and operation. The method discussed here with Figure 3.39 does not allow circulating current between converters 1 and 2. This type of control has a drawback: slower current transfer from one to the other bridge, due to the dead time; but it has the advantage that no additional passive component in the form of an interphase reactor is needed to limit the circulating current.

The other type of crossover control uses a circulating current between the thyristor bridges. It has the advantage of faster current crossover and high dynamic response, but it has the disadvantages of requiring additional interphase reactors and losses associated with them during the operation. The operational details of such an arrangement can be found in references.

3.10 CONVERTER SELECTION AND CHARACTERISTICS

The ratings of the converter and its power switches are derived from the motor and load specifications. Some approximate derivations are given in this section.

Let the maximum current allowed in the motor be $I_{\max}$. The rms value of the current in each device is then based on the fact that it is conducting for 120 electrical degrees in a cycle and that the current is flat. Such an assumption might not be strictly valid under discontinuous conduction, and, anyway, at those points the current will be very much lower than the rated value. The rms value of the current in the power device is

$$I_{\text{rms}} = \frac{I_{\max}}{\sqrt{3}} = 0.577I_{\max} \quad (3.120)$$

The voltage rating is the maximum line-to-line voltage of the ac mains.

$$V_t = \sqrt{2}V \quad (3.121)$$

I_1 is the fundamental rms component of the ac input current, which is derived as

$$I_1 = \frac{1}{\sqrt{2}} \cdot \frac{2\sqrt{3}}{\pi} \cdot I_{\max} = \frac{\sqrt{3}\sqrt{2}}{\pi} \cdot I_{\max} = 0.78I_{\max} \quad (3.122)$$

The output power of the converter is given by

$$P_o = V_a I_{\max} = \{1.35V \cos \alpha\} I_{\max} = 1.35V I_{\max} \cos \alpha \quad (3.123)$$

Neglecting losses in the converter, the input power equals the output power. Substituting for $I_{\max}$ in terms of the fundamental ac input current from equation (3.122) gives

$$P_i = P_o = 1.35V I_{\max} \cos \alpha = \sqrt{3}V I_1 \cos \alpha \quad (3.124)$$

This equation gives the real power in a balanced three-phase ac system with a power factor of $\cos \alpha$. Therefore, the angle α gives the power factor angle. Similarly, the reactive power is given by

$$Q_i = \sqrt{3}V I_1 \sin \alpha = 1.35V I_{\max} \sin \alpha \quad (3.125)$$

The input apparent power is

$$P_{VA} = \sqrt{(P_i)^2 + (Q_i)^2} = 1.35VI_{\max} = \sqrt{3}VI_1 \quad (3.126)$$

Because of the reactive power consumption of the phase-controlled converters, they are expensive to operate where the reactive power is to be paid for. They also generate harmonics, which may be unacceptable to power utilities beyond a certain limit. The above two features are the primary disadvantages of the phase-controlled converter-fed motor drives.

3.11 SIMULATION OF THE ONE-QUADRANT DC MOTOR DRIVE

The equations for various subsystems are derived and then assembled for computer simulation in this section. Key results are discussed. The simulation for either a two- or four-quadrant dc motor drive is very similar to the present development. In the present one-quadrant speed-controlled motor-drive simulation, it is assumed that the field current is constant in the constant torque mode and is varied through a three-phase controlled rectifier for the field-weakening mode of operation to provide constant power over a wide speed range.

3.11.1 The Motor Equations

The motor equations, including that of a simple load modeling, are given below:

$$V_a = R_a i_a + L_a \frac{di_a}{dt} + M i_f \omega_m \quad (3.127)$$

$$V_f = R_f i_f + L_f \frac{di_f}{dt} \quad (3.128)$$

$$M i_f i_a - T_l = J \frac{d\omega_m}{dt} + B \omega_m \quad (3.129)$$

These equations can be rearranged in the following form to facilitate their solution by numerical integration :

$$\frac{di_a}{dt} = -\frac{R_a}{L_a} i_a - \frac{M}{L_a} i_f \omega_m + \frac{V_a}{L_a} \quad (3.130)$$

$$\frac{di_f}{dt} = -\frac{R_f}{L_f} i_f + \frac{V_f}{L_f} \quad (3.131)$$

$$\frac{d\omega_m}{dt} = \frac{M}{J} i_f i_a - \frac{B}{J} \omega_m - \frac{T_l}{J} \quad (3.132)$$

Choose i_a , i_f , and ω_m as state variables and denote them as

$$x_1 = i_a \quad (3.133)$$

$$x_2 = i_f \quad (3.134)$$

$$x_3 = \omega_m \quad (3.135)$$